MATH 501: REAL ANALYSIS

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1. August 25, 2026

Real analysis will be developed through measure theory, Lebesgue integration, and the study of function spaces such as L^p spaces.

1.1. **Measures.** The goal of a measure is to assign a length, area, or volume to subsets of $\mathbb{R}$, $\mathbb{R}^2$, or, more generally, $\mathbb{R}^n$. Ideally, we would like a function

$$m\colon 2^{\mathbb{R}^n} \longrightarrow [0,\infty]$$

with the following properties.

(i) *Countable additivity:* if $E_1, E_2, \ldots \subseteq \mathbb{R}^n$ are pairwise disjoint, then

$$m\left(\bigcup_{j=1}^{\infty} E_j\right) = \sum_{j=1}^{\infty} m(E_j).$$

A weaker requirement is finite additivity, in which the identity is required only for finite pairwise disjoint families.

(ii) *Invariance under rigid motions:* if

$$\widetilde{E} = AE + b, \qquad A \in O(n), \quad b \in \mathbb{R}^n,$$

then $m(\widetilde{E}) = m(E)$. Here A represents a rotation or reflection, and b a translation.

(iii) *Normalization:*

$$m([0,1]^n) = 1.$$

Unfortunately, a measure with all these properties cannot be defined on every subset of $\mathbb{R}^n$.

Theorem 1 (Vitali)**.** *Assuming the axiom of choice, there is no countably additive, rigid-motion-invariant map*

$$m\colon 2^{\mathbb{R}^n} \longrightarrow [0,\infty]$$

satisfying $m([0,1]^n) = 1$.

Proof. It is enough to consider $n = 1$: if such a map existed on $\mathbb{R}^n$, then

$$\mu(E) = m\big(E \times [0,1]^{n-1}\big), \qquad E \subseteq \mathbb{R},$$

would have the required properties on $\mathbb{R}$. We therefore write m for this one-dimensional map.

First, $m(\{x\}) = 0$ for every $x \in \mathbb{R}$. Otherwise, translation invariance and countable additivity applied to a countably infinite set in $[0,1]$ would contradict $m([0,1]) = 1$. Consequently, $m([0,1)) = 1$.

Define an equivalence relation on $[0,1)$ by

$$x \sim y \quad \Longleftrightarrow \quad x - y \in \mathbb{Q}.$$

Using the axiom of choice, select one representative from each equivalence class and let V be the set of representatives. Thus, for every $x \in [0,1)$, there is a unique $y \in V$ such that $x - y \in \mathbb{Q}$.

For $t \in [0,1)$, define the translate of V modulo 1 by

$$\begin{aligned} V_t &= V + t \pmod 1 \\ &= \{x + t : x \in V,\ x + t < 1\} \\ &\quad \cup \{x + t - 1 : x \in V,\ x + t \geq 1\}. \end{aligned}$$

Finite additivity and translation invariance give

$$m(V_t) = m(V).$$

The family indexed by $t \in [0,1) \cap \mathbb{Q}$ is a partition of $[0,1)$. Indeed, if $x \in [0,1)$ and $y \in V$ represents the equivalence class of x, then either $t = x - y$ or $t = x - y + 1$ lies in $[0,1) \cap \mathbb{Q}$ and $x \in V_t$. On the other hand, if $V_t \cap V_{t'} \neq \varnothing$, the uniqueness of representatives implies $t = t'$.

Countable additivity now yields

$$1 = m([0,1)) = \sum_{t \in [0,1) \cap \mathbb{Q}} m(V_t) = \sum_{t \in [0,1) \cap \mathbb{Q}} m(V).$$

If $m(V) = 0$, the last sum is 0; if $m(V) > 0$, it is infinite. Either possibility is a contradiction. □

One might try to rescue the construction by requiring only finite additivity together with invariance and normalization. In dimensions at least three, even this is impossible.

Theorem 2 (Banach–Tarski)**.** *If $n \geq 3$, there is no finitely additive, rigid-motion-invariant map*

$$m\colon 2^{\mathbb{R}^n} \longrightarrow [0,\infty]$$

satisfying $m([0,1]^n) = 1$.

The resolution is to give up the requirement that a measure be defined on *all* subsets of $\mathbb{R}^n$. In fact, measure theory can be developed on a general underlying set X once an appropriate collection of measurable subsets has been chosen.

1.2. **Algebras and σ-algebras.**

Definition 1 (Algebra)**.** *An* algebra *on a set X is a nonempty collection $\mathcal{A} \subseteq 2^X$ such that*

(a) if $E_1, \ldots, E_N \in \mathcal{A}$, then $\bigcup_{j=1}^N E_j \in \mathcal{A}$;
(b) if $E \in \mathcal{A}$, then $E^c = X \setminus E \in \mathcal{A}$.

Definition 2 (σ-algebra)**.** *A* σ-algebra *on a set X is a nonempty collection $\mathcal{A} \subseteq 2^X$ such that*

(a) if $E_n \in \mathcal{A}$ for every $n \in \mathbb{N}$, then $\bigcup_{n=1}^\infty E_n \in \mathcal{A}$;
(b) if $E \in \mathcal{A}$, then $E^c \in \mathcal{A}$.

Because the collections in these definitions are nonempty and closed under complements, they contain both X and $\varnothing$. By De Morgan's laws, an algebra is closed under finite intersections and a σ-algebra is closed under countable intersections. For example,

$$\bigcap_{n=1}^{\infty} E_n = \left(\bigcup_{n=1}^{\infty} E_n^c \right)^c.$$

When an algebra is already known to be closed under finite set operations, it is enough to check closure under countable *disjoint* unions in order to prove that it is a σ-algebra. Given $E_n \in \mathcal{A}$, set

$$\widetilde{E}_1 = E_1, \qquad \widetilde{E}_n = E_n \setminus \bigcup_{j=1}^{n-1} E_j \quad (n \geq 2).$$

Then the sets $\widetilde{E}_n$ are pairwise disjoint and

$$\bigcup_{n=1}^{\infty} \widetilde{E}_n = \bigcup_{n=1}^{\infty} E_n.$$

Example 1. *The collection $\{\varnothing, X\}$ is the smallest σ-algebra on X.*

Example 2. *The power set 2^X is the largest σ-algebra on X.*

Example 3 (Countable–cocountable σ-algebra)**.** *The collection*

$$\mathcal{A} = \{E \subseteq X : E \text{ is countable or } E^c \text{ is countable}\}$$

is a σ-algebra on X.

A convenient way to specify a σ-algebra is to give a family of generators.

Definition 3 (Generated σ-algebra)**.** *Let $\mathcal{E} \subseteq 2^X$. The σ-algebra generated by $\mathcal{E}$, denoted $\mathcal{M}(\mathcal{E})$, is the smallest σ-algebra containing $\mathcal{E}$.*

This object exists because 2^X is a σ-algebra containing $\mathcal{E}$, and an arbitrary intersection of σ-algebras is again a σ-algebra. Explicitly,

$$\mathcal{M}(\mathcal{E}) = \bigcap \{\mathcal{A} : \mathcal{A} \text{ is a } \sigma\text{-algebra on } X \text{ and } \mathcal{E} \subseteq \mathcal{A}\}.$$

1.3. **Borel σ-algebras.**

Definition 4 (Borel σ-algebra)**.** *Let X be a metric space, or more generally a topological space. The* Borel σ-algebra $\mathcal{B}_X$ *is the σ-algebra generated by the open subsets of X.*

For example, every open subset of $\mathbb{R}$ is a union of open intervals. The Borel σ-algebra contains every open and closed set, every G_δ set (a countable intersection of open sets), every F_σ set (a countable union of closed sets), and all sets obtained by continuing such countable operations. These classes form the Borel hierarchy.

Proposition 1. *The Borel σ-algebra $\mathcal{B}_{\mathbb{R}}$ is generated by any one of the following collections:*

(a) $\{(a, b) : a < b\}$*;*

(b) $\{[a, b] : a < b\}$;
(c) $\{(-\infty, a] : a \in \mathbb{R}\}$.

Proof. Let the three collections above be $\mathcal{E}_1$, $\mathcal{E}_2$, and $\mathcal{E}_3$, respectively.

Every member of $\mathcal{E}_1$ is open, so $\mathcal{M}(\mathcal{E}_1) \subseteq \mathcal{B}_{\mathbb{R}}$. For the reverse inclusion, let $U \subseteq \mathbb{R}$ be open. For every $x \in U$, there is an $\varepsilon > 0$ such that

$$(x - \varepsilon, x + \varepsilon) \subseteq U.$$

By the density of $\mathbb{Q}$ in $\mathbb{R}$, we can choose $p, q \in \mathbb{Q}$ such that

$$x - \varepsilon < p < x < q < x + \varepsilon.$$

Thus every $x \in U$ lies in an interval (p, q) with rational endpoints that is contained in U. It follows that

$$U = \bigcup \{(p, q) : p, q \in \mathbb{Q},\ p < q,\ (p, q) \subseteq U\}.$$

The indexing collection is a subset of $\mathbb{Q}^2$, so it is countable. Therefore every open subset of $\mathbb{R}$ is a countable union of open intervals. In particular, every open set belongs to $\mathcal{M}(\mathcal{E}_1)$, proving

$$\mathcal{M}(\mathcal{E}_1) = \mathcal{B}_{\mathbb{R}}.$$

Every closed interval is Borel. In the other direction, for $a < b$,

$$(a, b) = \bigcup_{\substack{n \in \mathbb{N} \\ 2/n < b-a}} \left[a + \frac{1}{n}, b - \frac{1}{n}\right].$$

Thus $\mathcal{E}_1 \subseteq \mathcal{M}(\mathcal{E}_2)$, and consequently $\mathcal{M}(\mathcal{E}_2) = \mathcal{B}_{\mathbb{R}}$. Equivalently, the reverse containment can also be seen from

$$[a, b] = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, b + \frac{1}{n}\right).$$

Finally, every left half-line in $\mathcal{E}_3$ is closed and hence Borel. Conversely,

$$(-\infty, b) = \bigcup_{n=1}^{\infty} \left(-\infty, b - \frac{1}{n}\right]$$

and therefore

$$(a, b) = (-\infty, b) \cap (-\infty, a]^c \in \mathcal{M}(\mathcal{E}_3).$$

It follows that $\mathcal{E}_1 \subseteq \mathcal{M}(\mathcal{E}_3)$, so $\mathcal{M}(\mathcal{E}_3) = \mathcal{B}_{\mathbb{R}}$ as well. □

2. August 27, 2026

We continue our discussion of σ-algebras. Recall that a σ-algebra $\mathcal{M}$ on a set X contains $\varnothing$ and X, is closed under complements, and is closed under countable unions. If (X, d) is a metric space, then its Borel σ-algebra $\mathcal{B}_X$ is generated by the open subsets of X.

2.1. **Product σ-algebras.**

Let I be an index set, let X_i be a nonempty set for each $i \in I$, and set

$$X = \prod_{i \in I} X_i.$$

For each $i \in I$, let $\mathcal{M}_i$ be a σ-algebra on X_i, and let

$$\pi_i \colon X \longrightarrow X_i, \qquad \pi_i((x_j)_{j \in I}) = x_i,$$

be the coordinate projection.

Definition 5 (Product σ-algebra)**.** *The* product σ-algebra *on X is*

$$\bigotimes_{i \in I} \mathcal{M}_i = \mathcal{M}\left(\{\pi_i^{-1}(E) : i \in I,\ E \in \mathcal{M}_i\}\right).$$

Thus it is the smallest σ-algebra that makes every coordinate projection measurable.

The generating sets $\pi_i^{-1}(E)$ are called *cylinder sets*. They restrict the ith coordinate to E and leave all other coordinates unrestricted.

Lemma 1. *If I is countable, then $\bigotimes_{i \in I} \mathcal{M}_i$ is generated by the measurable rectangles*

$$\left\{ \prod_{i \in I} E_i : E_i \in \mathcal{M}_i \right\}.$$

Proof. Since I is countable, every measurable rectangle satisfies

$$\prod_{i \in I} E_i = \bigcap_{i \in I} \pi_i^{-1}(E_i) \in \bigotimes_{i \in I} \mathcal{M}_i.$$

Conversely, each cylinder set is a measurable rectangle. Indeed,

$$\pi_j^{-1}(E_j) = \prod_{i \in I} F_i, \qquad F_i = \begin{cases} E_j, & i = j, \\ X_i, & i \neq j. \end{cases}$$

The two collections therefore generate the same σ-algebra. □

The generators of the coordinate σ-algebras can be used directly to generate the product.

Proposition 2. *Suppose*

$$\mathcal{M}_i = \mathcal{M}(\mathcal{G}_i), \qquad \mathcal{G}_i \subseteq 2^{X_i},$$

for every $i \in I$.

(a) The product σ-algebra is generated by

$$\mathcal{F} = \{\pi_i^{-1}(G) : i \in I,\ G \in \mathcal{G}_i\}.$$

(b) If I is countable and $\mathcal{G}_i^* = \mathcal{G}_i \cup \{X_i\}$, then the product σ-algebra is generated by

$$\left\{ \prod_{i \in I} G_i : G_i \in \mathcal{G}_i^* \right\}.$$

Proof. For part (a), every member of $\mathcal{F}$ belongs to $\bigotimes_{i \in I} \mathcal{M}_i$, so

$$\mathcal{M}(\mathcal{F}) \subseteq \bigotimes_{i \in I} \mathcal{M}_i.$$

For the reverse inclusion, fix $i \in I$ and consider

$$\mathcal{A}_i = \{E \subseteq X_i : \pi_i^{-1}(E) \in \mathcal{M}(\mathcal{F})\}.$$

Preimages preserve complements and countable unions, so $\mathcal{A}_i$ is a σ-algebra. It contains $\mathcal{G}_i$ and hence contains $\mathcal{M}(\mathcal{G}_i) = \mathcal{M}_i$. Therefore every cylinder $\pi_i^{-1}(E)$ with $E \in \mathcal{M}_i$ belongs to $\mathcal{M}(\mathcal{F})$, which proves the reverse inclusion.

For part (b), each cylinder from part (a) is a rectangle whose other coordinates are X_j. Conversely, since I is countable, every rectangle in the displayed collection is the countable intersection of the sets $\pi_i^{-1}(G_i)$.

When $G_i = X_i$ this set is all of X; otherwise it is a cylinder from part (a). The conclusion follows. □

2.2. **Products of Borel σ-algebras.** Let (X_i, d_i), $i = 1, \dots, n$, be metric spaces. On

$$X = \prod_{i=1}^{n} X_i$$

define the maximum metric

$$d(x, y) = \max_{1 \le i \le n} d_i(x_i, y_i).$$

Proposition 3. *With the notation above,*

$$\bigotimes_{i=1}^{n} \mathcal{B}_{X_i} \subseteq \mathcal{B}_X.$$

If every (X_i, d_i) is separable, then equality holds:

$$\bigotimes_{i=1}^{n} \mathcal{B}_{X_i} = \mathcal{B}_X.$$

Proof. Each coordinate projection $\pi_i \colon X \to X_i$ is continuous. If $U_i \subseteq X_i$ is open, then $\pi_i^{-1}(U_i)$ is open in X. Since $\mathcal{B}_{X_i}$ is generated by the open subsets of X_i, Proposition 2 gives

$$\bigotimes_{i=1}^{n} \mathcal{B}_{X_i} \subseteq \mathcal{B}_X.$$

Now suppose that each X_i is separable, and choose a countable dense subset $D_i \subseteq X_i$. The collection

$$\mathcal{G}_i = \{B_{d_i}(p, q) : p \in D_i,\ q \in \mathbb{Q},\ q > 0\}$$

is a countable basis for X_i and hence generates $\mathcal{B}_{X_i}$. The finite product $D = \prod_{i=1}^{n} D_i$ is countable and dense in X. Moreover,

$$B_d(x, q) = \prod_{i=1}^{n} B_{d_i}(x_i, q).$$

Thus every ball in X whose center lies in D and whose radius is a positive rational belongs to $\bigotimes_{i=1}^{n} \mathcal{B}_{X_i}$. These balls form a countable basis for X, so every open subset of X is a countable union of such

balls. Hence

$$\mathcal{B}_X \subseteq \bigotimes_{i=1}^{n} \mathcal{B}_{X_i},$$

which completes the proof. □

Corollary 1. *For every $n \in \mathbb{N}$,*

$$\mathcal{B}_{\mathbb{R}^n} = \bigotimes_{i=1}^{n} \mathcal{B}_{\mathbb{R}}.$$

For example, $\mathcal{B}_{\mathbb{R}^n}$ is generated by each of the following collections:

- open rectangles $\prod_{i=1}^{n}(a_i, b_i)$;
- closed rectangles $\prod_{i=1}^{n}[a_i, b_i]$; and
- lower orthants $\prod_{i=1}^{n}(-\infty, a_i]$.

2.3. **Measures.**

Definition 6 (Measure)**.** *Let $\mathcal{M}$ be a σ-algebra on a set X. A function*

$$\mu \colon \mathcal{M} \longrightarrow [0, \infty]$$

is a measure *if*

(i) $\mu(\varnothing) = 0$*;*

(ii) whenever $E_1, E_2, \ldots \in \mathcal{M}$ are pairwise disjoint,

$$\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu(E_i).$$

The second property is called countable additivity*. If it is required only for finite pairwise disjoint families, then μ is called finitely additive.*

The sets in $\mathcal{M}$ are called *measurable sets*, and the triple $(X, \mathcal{M}, \mu)$ is called a *measure space*. A measure μ is

- *finite* if $\mu(X) < \infty$;
- *σ-finite* if $X = \bigcup_{i=1}^{\infty} E_i$ for some $E_i \in \mathcal{M}$ with $\mu(E_i) < \infty$ for every i.

A set $N \in \mathcal{M}$ with $\mu(N) = 0$ is called a *null set*. A statement is said to hold *almost everywhere*, abbreviated a.e., if it holds outside some null set.

2.4. **Examples of measures.**

Example 4 (Weighted counting measure)**.** *Let X be nonempty, let $\mathcal{M} = 2^X$, and let $f\colon X \to [0,\infty]$. Define*

$$\mu(E) = \sum_{x\in E} f(x) := \sup\left\{\sum_{x\in F} f(x) : F \subseteq E,\ F \text{ finite}\right\}.$$

Then μ is a measure on 2^X.

If $f(x) = 1$ for every $x \in X$, then μ is the counting measure*,*

$$\mu(E) = \#E.$$

If $x_0 \in X$ and

$$f(x) = \begin{cases} 1, & x = x_0, \\ 0, & x \neq x_0, \end{cases}$$

then

$$\mu(E) = \begin{cases} 1, & x_0 \in E, \\ 0, & x_0 \notin E. \end{cases}$$

This is the point mass*, or* Dirac measure*, at x_0.*

Example 5 (Countable–cocountable measure)**.** *Let X be uncountable, and let $\mathcal{M}$ be the countable–cocountable σ-algebra. Then*

$$\mu(E) = \begin{cases} 0, & E \text{ is countable}, \\ 1, & E^c \text{ is countable} \end{cases}$$

defines a measure on $(X, \mathcal{M})$.

Example 6 (Finite additivity is not enough)**.** *Let $X = \mathbb{N}$ and $\mathcal{M} = 2^{\mathbb{N}}$. Define*

$$\nu(E) = \begin{cases} 0, & E \text{ is finite}, \\ \infty, & E \text{ is infinite}. \end{cases}$$

Then ν is finitely additive. It is not a measure, because

$$\nu\left(\bigcup_{n=1}^{\infty}\{n\}\right) = \infty \qquad \text{but} \qquad \sum_{n=1}^{\infty}\nu(\{n\}) = 0.$$

2.5. **Basic properties of measures.**

Theorem 3. *Let $(X, \mathcal{M}, \mu)$ be a measure space.*

(a) Monotonicity: *if $E, F \in \mathcal{M}$ and $E \subseteq F$, then*

$$\mu(E) \leq \mu(F).$$

(b) Countable subadditivity: *if $E_1, E_2, \ldots \in \mathcal{M}$, then*

$$\mu\left(\bigcup_{i=1}^{\infty} E_i\right) \leq \sum_{i=1}^{\infty} \mu(E_i).$$

(c) Continuity from below: *if*

$$E_1 \subseteq E_2 \subseteq \cdots, \qquad E_i \in \mathcal{M},$$

then

$$\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \lim_{i\to\infty} \mu(E_i).$$

(d) Continuity from above: *if*

$$E_1 \supseteq E_2 \supseteq \cdots, \qquad E_i \in \mathcal{M}, \qquad \mu(E_1) < \infty,$$

then

$$\mu\left(\bigcap_{i=1}^{\infty} E_i\right) = \lim_{i\to\infty} \mu(E_i).$$

Proof. For part (a), write F as the disjoint union

$$F = E \cup (F \setminus E).$$

Countable additivity gives

$$\mu(F) = \mu(E) + \mu(F \setminus E) \geq \mu(E).$$

For part (b), define

$$F_1 = E_1, \qquad F_k = E_k \setminus \bigcup_{j=1}^{k-1} E_j \quad (k \geq 2).$$

The sets F_k are pairwise disjoint, $F_k \subseteq E_k$, and

$$\bigcup_{k=1}^{\infty} F_k = \bigcup_{k=1}^{\infty} E_k.$$

Therefore countable additivity and part (a) imply

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu(F_k) \leq \sum_{k=1}^{\infty} \mu(E_k).$$

For part (c), set

$$F_1 = E_1, \qquad F_k = E_k \setminus E_{k-1} \quad (k \geq 2).$$

The sets F_k are pairwise disjoint, and for each N,

$$E_N = \bigcup_{k=1}^{N} F_k, \qquad \bigcup_{i=1}^{\infty} E_i = \bigcup_{k=1}^{\infty} F_k.$$

It follows that

$$\begin{aligned} \mu\left(\bigcup_{i=1}^{\infty} E_i\right) &= \sum_{k=1}^{\infty} \mu(F_k) \\ &= \lim_{N\to\infty} \sum_{k=1}^{N} \mu(F_k) \\ &= \lim_{N\to\infty} \mu(E_N). \end{aligned}$$

For part (d), define

$$F_i = E_1 \setminus E_i.$$

Since E_i decreases, F_i increases, and

$$\bigcup_{i=1}^{\infty} F_i = E_1 \setminus \bigcap_{i=1}^{\infty} E_i.$$

By part (c),

$$\mu\left(E_1 \setminus \bigcap_{i=1}^{\infty} E_i\right) = \lim_{i\to\infty} \mu(E_1 \setminus E_i).$$

Because $E_i \subseteq E_1$ and $\mu(E_1) < \infty$, finite additivity gives

$$\mu(E_1 \setminus E_i) = \mu(E_1) - \mu(E_i)$$

and

$$\mu\left(E_1 \setminus \bigcap_{i=1}^{\infty} E_i\right) = \mu(E_1) - \mu\left(\bigcap_{i=1}^{\infty} E_i\right).$$

Substituting these identities into the preceding limit and cancelling the finite quantity $\mu(E_1)$ yields

$$\mu\left(\bigcap_{i=1}^{\infty} E_i\right) = \lim_{i\to\infty} \mu(E_i).$$

□

Remark 1 (Necessity of the finite-measure hypothesis). The assumption $\mu(E_1) < \infty$ in Theorem 3(d) cannot be omitted. Let

$$X = \mathbb{N}, \qquad \mathcal{M} = 2^{\mathbb{N}},$$

let μ be counting measure, and define

$$E_i = \{i, i+1, i+2, \ldots\}.$$

Then

$$E_1 \supseteq E_2 \supseteq E_3 \supseteq \cdots, \qquad \bigcap_{i=1}^{\infty} E_i = \varnothing.$$

Indeed, for any $n \in \mathbb{N}$, we have $n \notin E_i$ whenever $i > n$. Consequently,

$$\mu\left(\bigcap_{i=1}^{\infty} E_i\right) = \mu(\varnothing) = 0.$$

On the other hand, every E_i is infinite, so

$$\mu(E_i) = \infty \quad \text{for every } i, \qquad \lim_{i\to\infty} \mu(E_i) = \infty.$$

Thus

$$\mu\left(\bigcap_{i=1}^{\infty} E_i\right) = 0 \neq \infty = \lim_{i\to\infty} \mu(E_i).$$

This does not contradict part (d), because $\mu(E_1) = \mu(\mathbb{N}) = \infty$.

3. September 1, 2026

Recall that a measure on a σ-algebra $\mathcal{M}$ over a set X is a map

$$\mu \colon \mathcal{M} \longrightarrow [0, \infty]$$

such that $\mu(\varnothing) = 0$ and

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

whenever $A_1, A_2, \ldots \in \mathcal{M}$ are pairwise disjoint. For Lebesgue measure m on $\mathbb{R}$, the normalization is $m([0,1]) = 1$.

3.1. **Complete measure spaces.**

Definition 7 (Complete measure)**.** *A measure μ on a σ-algebra $\mathcal{M}$ over X is* complete *if every subset of a null set is measurable. That is, if $E \subseteq N$ for some $N \in \mathcal{M}$ with $\mu(N) = 0$, then $E \in \mathcal{M}$.*

By monotonicity, every such set E also satisfies

$$0 \leq \mu(E) \leq \mu(N) = 0,$$

so $\mu(E) = 0$.

Every measure space has a canonical completion obtained by adjoining all subsets of its null sets.

Proposition 4 (Completion of a measure)**.** *Let $(X, \mathcal{M}, \mu)$ be a measure space, and let*

$$\mathcal{N} = \{N \in \mathcal{M} : \mu(N) = 0\}.$$

Define

$$\overline{\mathcal{M}} = \{E \cup F : E \in \mathcal{M},\ F \subseteq N \text{ for some } N \in \mathcal{N}\}.$$

Then the following statements hold.

(i) The collection $\overline{\mathcal{M}}$ is a σ-algebra on X containing $\mathcal{M}$.

(ii) There is a unique measure $\overline{\mu}$ on $\overline{\mathcal{M}}$ extending μ. It is given by

$$\overline{\mu}(E \cup F) = \mu(E),$$

where $E \in \mathcal{M}$ and F is contained in a member of $\mathcal{N}$. Moreover, $\overline{\mu}$ is complete.

Proof. The collections $\mathcal{M}$ and $\mathcal{N}$ are closed under countable unions. For $\mathcal{N}$, this follows from countable subadditivity: if $N_i \in \mathcal{N}$, then

$$0 \leq \mu\left(\bigcup_{i=1}^{\infty} N_i\right) \leq \sum_{i=1}^{\infty} \mu(N_i) = 0.$$

Hence $\overline{\mathcal{M}}$ is closed under countable unions. Indeed, if $E_i \in \mathcal{M}$ and $F_i \subseteq N_i \in \mathcal{N}$, then

$$\bigcup_{i=1}^{\infty}(E_i \cup F_i) = \left(\bigcup_{i=1}^{\infty} E_i\right) \cup \left(\bigcup_{i=1}^{\infty} F_i\right),$$

and $\bigcup_i F_i \subseteq \bigcup_i N_i \in \mathcal{N}$.

To check closure under complements, let $E \in \mathcal{M}$ and $F \subseteq N \in \mathcal{N}$. Replacing N by $N \setminus E$ and F by $F \setminus E$ does not change $E \cup F$, so we may assume that $E \cap N = \varnothing$. In that case,

$$(E \cup F)^c = (E \cup N)^c \cup (N \setminus F).$$

The first set on the right belongs to $\mathcal{M}$, and the second is contained in the null set N. Thus $(E \cup F)^c$ belongs to $\overline{\mathcal{M}}$. Since $\varnothing \in \overline{\mathcal{M}}$, this proves that $\overline{\mathcal{M}}$ is a σ-algebra. Taking $F = \varnothing$ shows that $\mathcal{M} \subseteq \overline{\mathcal{M}}$.

We next show that the formula for $\overline{\mu}$ is well defined. Suppose

$$E_1 \cup F_1 = E_2 \cup F_2, \qquad E_i \in \mathcal{M}, \quad F_i \subseteq N_i \in \mathcal{N}.$$

Then $E_1 \subseteq E_2 \cup N_2$, and therefore

$$\mu(E_1) \leq \mu(E_2 \cup N_2) \leq \mu(E_2) + \mu(N_2) = \mu(E_2).$$

By symmetry, $\mu(E_2) \leq \mu(E_1)$, so $\mu(E_1) = \mu(E_2)$.

Clearly $\overline{\mu}(\varnothing) = 0$. Suppose that $A_i = E_i \cup F_i$ are pairwise disjoint members of $\overline{\mathcal{M}}$, where $E_i \in \mathcal{M}$ and $F_i \subseteq N_i \in \mathcal{N}$. The sets E_i are pairwise disjoint, so

$$\begin{aligned}\overline{\mu}\left(\bigcup_{i=1}^{\infty} A_i\right) &= \mu\left(\bigcup_{i=1}^{\infty} E_i\right) \\ &= \sum_{i=1}^{\infty} \mu(E_i)\end{aligned}$$

$$= \sum_{i=1}^{\infty} \overline{\mu}(A_i).$$

Thus $\overline{\mu}$ is a measure, and it extends μ.

If $A = E \cup F \in \overline{\mathcal{M}}$ and $\overline{\mu}(A) = 0$, then $\mu(E) = 0$. Since $A \subseteq E \cup N$ and $E \cup N$ is a measurable null set, every subset of A belongs to $\overline{\mathcal{M}}$. Hence $\overline{\mu}$ is complete.

Finally, let ν be any measure on $\overline{\mathcal{M}}$ that extends μ. If $A = E \cup F$, with $E \in \mathcal{M}$ and $F \subseteq N \in \mathcal{N}$, then

$$E \subseteq A \subseteq E \cup N.$$

Monotonicity and the fact that ν extends μ give

$$\mu(E) = \nu(E) \leq \nu(A) \leq \nu(E \cup N) = \mu(E \cup N) = \mu(E).$$

Thus $\nu(A) = \mu(E) = \overline{\mu}(A)$, which proves uniqueness. □

The measure $\overline{\mu}$ is called the *completion* of μ.

3.2. **Outer measures.** The goal is to construct measures from simple geometric pieces. For example, to estimate the area of a set $E \subseteq \mathbb{R}^2$, one can cover E by squares of side length s and consider

$$\inf_{s>0} \big(\text{number of squares in the cover}\big) s^2.$$

More generally, one covers a set by countably many simple pieces and takes the infimum of the sums of their assigned sizes.

A related guiding principle is that the "reasonable" subsets $E \subseteq [0,1]^2$ should be those for which outer and inner area agree. If the unit square has area 1, its inner estimate for E is

$$1 - m^*\big([0,1]^2 \setminus E\big).$$

Definition 8 (Outer measure)**.** *An* outer measure *on a set X is a map*

$$\mu^* \colon 2^X \longrightarrow [0, \infty]$$

satisfying the following properties.

(i) $\mu^*(\varnothing) = 0$.

(ii) If $A \subseteq B \subseteq X$*, then* $\mu^*(A) \leq \mu^*(B)$.

(iii) For every sequence $A_1, A_2, \ldots \subseteq X$*,*

$$\mu^*\left(\bigcup_{j=1}^{\infty} A_j\right) \leq \sum_{j=1}^{\infty} \mu^*(A_j).$$

Unlike a measure, an outer measure is defined on every subset of X; its countable subadditivity therefore applies without a measurability assumption.

Proposition 5 (Outer measure from coverings)**.** *Let* $\mathcal{E} \subseteq 2^X$ *contain* $\varnothing$ *and* X*, and let*

$$\rho\colon \mathcal{E} \longrightarrow [0, \infty]$$

satisfy $\rho(\varnothing) = 0$*. For* $A \subseteq X$*, define*

$$\mu^*(A) = \inf\left\{\sum_{j=1}^{\infty} \rho(E_j) : E_j \in \mathcal{E}, \quad A \subseteq \bigcup_{j=1}^{\infty} E_j\right\}.$$

Then μ^* *is an outer measure on* X*.*

Remark 2. Even when $E \in \mathcal{E}$, it need not be true that $\mu^*(E) = \rho(E)$. A more efficient countable cover may have smaller total ρ-size than the single-set cover by E.

Example 7. *Let* $X = \mathbb{R}$ *and take*

$$\mathcal{E} = \{\varnothing, \mathbb{R}\} \cup \{[a, b] : a, b \in \mathbb{R},\ a \leq b\}.$$

Define

$$\rho(\varnothing) = 0, \qquad \rho(\mathbb{R}) = \infty, \qquad \rho([a, b]) = (b - a)^2.$$

Although $\rho([0, 1]) = 1$*, the induced outer measure satisfies* $\mu^*([0, 1]) = 0$*. Indeed, for every* $n \in \mathbb{N}$*,*

$$[0, 1] = \bigcup_{j=0}^{n-1} \left[\frac{j}{n}, \frac{j+1}{n}\right],$$

and the total ρ*-size of this cover is*

$$\sum_{j=0}^{n-1} \left(\frac{1}{n}\right)^2 = \frac{1}{n}.$$

Hence $0 \leq \mu^*([0, 1]) \leq 1/n$ *for every* n*, which gives the claim.*

Proof of the proposition. The defining collection of sums is nonempty because any set $A \subseteq X$ is covered by the sequence $X, \varnothing, \varnothing, \ldots$. Thus $\mu^*(A)$ is well defined as an element of $[0, \infty]$.

The sequence $\varnothing, \varnothing, \ldots$ covers $\varnothing$, so

$$0 \leq \mu^*(\varnothing) \leq \sum_{j=1}^{\infty} \rho(\varnothing) = 0.$$

If $A \subseteq B$, then every countable cover of B by members of $\mathcal{E}$ is also a cover of A. Taking infima over the respective collections of covers gives $\mu^*(A) \leq \mu^*(B)$.

It remains to prove countable subadditivity. Let $A_j \subseteq X$ for $j \in \mathbb{N}$. If $\sum_j \mu^*(A_j) = \infty$, there is nothing to prove. Otherwise, fix $\varepsilon > 0$. For every j, choose $E_{j,k} \in \mathcal{E}$, $k \in \mathbb{N}$, such that

$$A_j \subseteq \bigcup_{k=1}^{\infty} E_{j,k} \qquad \text{and} \qquad \sum_{k=1}^{\infty} \rho(E_{j,k}) \leq \mu^*(A_j) + \varepsilon 2^{-j}.$$

The doubly indexed collection $\{E_{j,k} : j, k \in \mathbb{N}\}$ is countable and covers $\bigcup_j A_j$. Consequently,

$$\begin{aligned} \mu^*\left(\bigcup_{j=1}^{\infty} A_j\right) &\leq \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \rho(E_{j,k}) \\ &\leq \sum_{j=1}^{\infty} \left(\mu^*(A_j) + \varepsilon 2^{-j}\right) \\ &= \sum_{j=1}^{\infty} \mu^*(A_j) + \varepsilon. \end{aligned}$$

Letting $\varepsilon \downarrow 0$ proves the result. □

Definition 9 (Measurability with respect to an outer measure)**.** *Let μ^* be an outer measure on X. A set $A \subseteq X$ is* μ^*-measurable *if, for every $E \subseteq X$,*

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

Countable subadditivity always gives

$$\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c),$$

so the content of the measurability criterion is the reverse inequality. When $\mu^*(E) < \infty$, the quantity

$$\mu^*(E) - \mu^*(E \cap A^c)$$

can be viewed as the inner measure of $E \cap A$ relative to E.

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Recall that if μ^* is an outer measure on X, a set $A \subseteq X$ is μ^*-measurable if

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

for every $E \subseteq X$. By countable subadditivity, it is enough to prove the inequality

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

4.1. **The Carathéodory construction.**

Theorem 4 (Carathéodory)**.** *Let μ^* be an outer measure on X, and set*

$$\mathcal{M}^* = \{A \subseteq X : A \text{ is } \mu^*\text{-measurable}\}.$$

Then $\mathcal{M}^*$ is a σ-algebra, and

$$\mu = \mu^*|_{\mathcal{M}^*}$$

is a complete measure on $\mathcal{M}^*$.

Proof. We first show that $\mathcal{M}^*$ is an algebra. The empty set is measurable because

$$\mu^*(E) = \mu^*(E \cap \varnothing) + \mu^*(E \cap X)$$

for every $E \subseteq X$. The defining condition is symmetric in A and A^c, so

$$A \in \mathcal{M}^* \quad \Longrightarrow \quad A^c \in \mathcal{M}^*.$$

Now let $A, B \in \mathcal{M}^*$ and fix $E \subseteq X$. Applying the measurability criterion first to A and then to B gives

$$\begin{aligned}\mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c) \\ &= \mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) \\ &\quad + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c).\end{aligned}$$

The first three sets on the last two lines have union $E \cap (A \cup B)$. Therefore countable subadditivity implies

$$\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c).$$

Thus $A \cup B \in \mathcal{M}^*$, and $\mathcal{M}^*$ is an algebra.

This argument also establishes finite additivity on disjoint measurable sets. Indeed, if $A, B \in \mathcal{M}^*$ are disjoint, apply the measurability of A with $E = A \cup B$ to obtain

$$\mu^*(A \cup B) = \mu^*((A \cup B) \cap A) + \mu^*((A \cup B) \cap A^c) = \mu^*(A) + \mu^*(B).$$

To prove closure under countable unions, first let $A_1, A_2, \ldots \in \mathcal{M}^*$ be pairwise disjoint, and set

$$B = \bigcup_{i=1}^{\infty} A_i, \qquad B_n = \bigcup_{i=1}^{n} A_i.$$

Since $\mathcal{M}^*$ is an algebra, $B_n \in \mathcal{M}^*$. Repeatedly applying the measurability criterion and using the disjointness of the A_i gives, for every $E \subseteq X$,

$$\mu^*(E) = \sum_{i=1}^{n} \mu^*(E \cap A_i) + \mu^*(E \cap B_n^c).$$

Because $B^c \subseteq B_n^c$, monotonicity yields

$$\mu^*(E) \geq \sum_{i=1}^{n} \mu^*(E \cap A_i) + \mu^*(E \cap B^c).$$

Letting $n \to \infty$ and then using countable subadditivity, we find

$$\mu^*(E) \geq \sum_{i=1}^{\infty} \mu^*(E \cap A_i) + \mu^*(E \cap B^c)$$

$$\geq \mu^*\left(\bigcup_{i=1}^{\infty}(E\cap A_i)\right)+\mu^*(E\cap B^c)$$
$$=\mu^*(E\cap B)+\mu^*(E\cap B^c).$$

Hence $B\in\mathcal{M}^*$. An arbitrary countable union can be disjointified by replacing A_i with

$$A_i\setminus\bigcup_{j=1}^{i-1}A_j.$$

It follows that $\mathcal{M}^*$ is a σ-algebra.

The same computation proves countable additivity. If the measurable sets A_i are pairwise disjoint and $B=\bigcup_i A_i$, then countable subadditivity gives

$$\mu^*(B)\leq\sum_{i=1}^{\infty}\mu^*(A_i).$$

On the other hand, for every n,

$$\mu^*(B)=\sum_{i=1}^{n}\mu^*(A_i)+\mu^*(B\setminus B_n)\geq\sum_{i=1}^{n}\mu^*(A_i).$$

Letting $n\to\infty$ proves

$$\mu^*(B)=\sum_{i=1}^{\infty}\mu^*(A_i).$$

Thus $\mu=\mu^*|_{\mathcal{M}^*}$ is a measure.

Finally, suppose that $A\subseteq X$ and $\mu^*(A)=0$. For every $E\subseteq X$, subadditivity and monotonicity give

$$\mu^*(E\cap A)+\mu^*(E\cap A^c)\leq 0+\mu^*(E)=\mu^*(E).$$

Hence $A\in\mathcal{M}^*$. In particular, if $N\in\mathcal{M}^*$ satisfies $\mu(N)=0$, then every subset $A\subseteq N$ has outer measure zero and therefore belongs to $\mathcal{M}^*$. This proves that μ is complete. □

4.2. **Premeasures.** The covering construction from the preceding lecture begins with a set function ρ defined on a family of elementary sets.

The example $\rho([a,b]) = (b-a)^2$ showed that the resulting outer measure may fail to agree with ρ, even on elementary sets. To prevent this collapse, we require the initial set function to behave like a measure.

Definition 10 (Premeasure)**.** *Let $\mathcal{A}$ be an algebra on X. A map*

$$\mu_0 \colon \mathcal{A} \longrightarrow [0,\infty]$$

is a premeasure *if*

(i) $\mu_0(\varnothing) = 0$*;*

(ii) if $A_1, A_2, \ldots \in \mathcal{A}$ are pairwise disjoint and $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$, then

$$\mu_0\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu_0(A_i).$$

Thus a premeasure has the same additivity property as a measure, but its domain is only required to be an algebra, rather than a σ-algebra.

Example 8 (Half-open intervals)**.** *Let $X = \mathbb{R}$. Consider intervals of the forms*

$$(a,b], \qquad (-\infty,b], \qquad (a,\infty),$$

together with $\varnothing$, where $a, b \in \mathbb{R}$ and $a < b$. The collection $\mathcal{A}$ of finite unions of such intervals is an algebra. Moreover,

$$\mathcal{M}(\mathcal{A}) = \mathcal{B}_{\mathbb{R}}.$$

This algebra will later be used to construct Lebesgue measure.

The next proposition identifies the hypotheses under which the outer measure construction preserves the initial set function.

Proposition 6 (Extension of a premeasure)**.** *Let μ_0 be a premeasure on an algebra $\mathcal{A}$ over X. For every $E \subseteq X$, define*

$$\mu^*(E) = \inf\left\{\sum_{i=1}^{\infty} \mu_0(A_i) : A_i \in \mathcal{A}, \quad E \subseteq \bigcup_{i=1}^{\infty} A_i\right\}.$$

Then the following statements hold.

(i) $\mu^*|_{\mathcal{A}} = \mu_0$*.*

(ii) Every member of $\mathcal{A}$ is μ^*-measurable.

Consequently, $\mathcal{M}(\mathcal{A}) \subseteq \mathcal{M}^*$, and the measure $\mu^*|_{\mathcal{M}(\mathcal{A})}$ extends μ_0.

Proof. Let $A \in \mathcal{A}$. The sequence $A, \varnothing, \varnothing, \ldots$ is a cover of A, so

$$\mu^*(A) \leq \mu_0(A).$$

Conversely, suppose that $A_i \in \mathcal{A}$ and $A \subseteq \bigcup_{i=1}^{\infty} A_i$. Define

$$B_1 = A \cap A_1, \qquad B_k = A \cap \left(A_k \setminus \bigcup_{i=1}^{k-1} A_i \right) \quad (k \geq 2).$$

Then the sets B_k are pairwise disjoint members of $\mathcal{A}$, and

$$A = \bigcup_{k=1}^{\infty} B_k.$$

Since $B_k \subseteq A_k$, monotonicity of the premeasure gives

$$\mu_0(A) = \sum_{k=1}^{\infty} \mu_0(B_k) \leq \sum_{k=1}^{\infty} \mu_0(A_k).$$

Taking the infimum over all such covers yields $\mu_0(A) \leq \mu^*(A)$. Therefore $\mu^*(A) = \mu_0(A)$, proving part (i).

For part (ii), fix $A \in \mathcal{A}$ and $E \subseteq X$. If $\mu^*(E) = \infty$, the desired measurability identity follows at once from countable subadditivity. Suppose instead that $\mu^*(E) < \infty$, and let $\varepsilon > 0$. Choose $A_1, A_2, \ldots \in \mathcal{A}$ such that

$$E \subseteq \bigcup_{i=1}^{\infty} A_i \qquad \text{and} \qquad \mu^*(E) + \varepsilon \geq \sum_{i=1}^{\infty} \mu_0(A_i).$$

Since μ_0 is finitely additive,

$$\begin{aligned} \sum_{i=1}^{\infty} \mu_0(A_i) &= \sum_{i=1}^{\infty} \big(\mu_0(A_i \cap A) + \mu_0(A_i \cap A^c) \big) \\ &\geq \mu^*(E \cap A) + \mu^*(E \cap A^c). \end{aligned}$$

Hence

$$\mu^*(E) + \varepsilon \geq \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

Letting $\varepsilon \downarrow 0$ proves that A is μ^*-measurable.

By Theorem 4, $\mathcal{M}^*$ is a σ-algebra. Since it contains $\mathcal{A}$, it also contains $\mathcal{M}(\mathcal{A})$. Part (i) then shows that the restriction of μ^* to $\mathcal{M}(\mathcal{A})$ extends μ_0. □

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We continue the construction of measures from premeasures. Recall that if μ_0 is a premeasure on an algebra $\mathcal{A}$, then

$$\mu^*(E) = \inf\left\{\sum_{j=1}^{\infty} \mu_0(A_j) : A_j \in \mathcal{A}, \quad E \subseteq \bigcup_{j=1}^{\infty} A_j\right\}$$

defines an outer measure. Every member of $\mathcal{A}$ is μ^*-measurable, and

$$\mu = \mu^*|_{\mathcal{M}(\mathcal{A})}$$

is a measure extending μ_0.

5.1. **Uniqueness of the extension.**

The extension is unique on sets of finite measure. Under a σ-finiteness assumption, it is unique everywhere.

Theorem 5 (Uniqueness of the premeasure extension)**.** *Let μ_0 be a premeasure on an algebra $\mathcal{A}$, and let μ be the measure on $\mathcal{M}(\mathcal{A})$ obtained from the induced outer measure. Suppose that ν is another measure on $\mathcal{M}(\mathcal{A})$ extending μ_0.*

(a) If $E \in \mathcal{M}(\mathcal{A})$ and $\mu(E) < \infty$, then $\nu(E) = \mu(E)$.
(b) If μ_0 is σ-finite, then $\nu = \mu$ on $\mathcal{M}(\mathcal{A})$.

Proof. We divide the proof of part (a) into three steps.

First, let $E \in \mathcal{M}(\mathcal{A})$ and suppose that $E \subseteq \bigcup_{j=1}^{\infty} A_j$, where $A_j \in \mathcal{A}$. Since ν extends μ_0, countable subadditivity gives

$$\nu(E) \le \nu\left(\bigcup_{j=1}^{\infty} A_j\right) \le \sum_{j=1}^{\infty} \nu(A_j) = \sum_{j=1}^{\infty} \mu_0(A_j).$$

Taking the infimum over all such covers yields

$$\nu(E) \le \mu^*(E) = \mu(E). \tag{1}$$

Second, suppose that

$$A = \bigcup_{j=1}^{\infty} A_j, \qquad A_j \in \mathcal{A}.$$

Set $B_n = \bigcup_{j=1}^{n} A_j$. Then $B_n \in \mathcal{A}$ and $B_n \uparrow A$. Continuity from below, together with the fact that μ and ν both extend μ_0, gives

$$\nu(A) = \lim_{n\to\infty} \nu(B_n) = \lim_{n\to\infty} \mu(B_n) = \mu(A).$$

Third, let $E \in \mathcal{M}(\mathcal{A})$ satisfy $\mu(E) < \infty$, and fix $\varepsilon > 0$. By the definition of μ^*, there are $A_j \in \mathcal{A}$ such that, with $A = \bigcup_{j=1}^{\infty} A_j$,

$$E \subseteq A, \qquad \mu(A) \le \sum_{j=1}^{\infty} \mu_0(A_j) < \mu(E) + \varepsilon.$$

In particular, $\mu(A \setminus E) < \varepsilon$. By the second step and (1),

$$\begin{aligned} \mu(E) &\le \mu(A) = \nu(A) \\ &= \nu(E) + \nu(A \setminus E) \\ &\le \nu(E) + \mu(A \setminus E) < \nu(E) + \varepsilon. \end{aligned}$$

Letting $\varepsilon \downarrow 0$ shows that $\mu(E) \le \nu(E)$. Together with (1), this proves part (a).

For part (b), σ-finiteness provides sets $A_j \in \mathcal{A}$ such that

$$X = \bigcup_{j=1}^{\infty} A_j, \qquad \mu_0(A_j) < \infty.$$

By replacing A_j with $A_j \setminus \bigcup_{i=1}^{j-1} A_i$, we may assume that the sets are pairwise disjoint. If $E \in \mathcal{M}(\mathcal{A})$, then $\mu(E \cap A_j) \le \mu_0(A_j) < \infty$, so part (a) applies to each $E \cap A_j$. Therefore

$$\mu(E) = \sum_{j=1}^{\infty} \mu(E \cap A_j) = \sum_{j=1}^{\infty} \nu(E \cap A_j) = \nu(E).$$

□

5.2. **Lebesgue–Stieltjes measures on $\mathbb{R}$.** Let μ be a finite Borel measure on $\mathbb{R}$. Its *distribution function* is

$$F(x) = \mu((-\infty, x]), \qquad x \in \mathbb{R}. \tag{2}$$

This function records the measure of every left half-line.

Proposition 7. *The distribution function F has the following properties.*

(i) It is increasing.
(ii) It is right-continuous.
(iii) For every $a < b$,

$$\mu((a, b]) = F(b) - F(a).$$

Proof. If $x \le y$, then $(-\infty, x] \subseteq (-\infty, y]$, so monotonicity of μ gives $F(x) \le F(y)$.

If $x_n \downarrow x$, then

$$(-\infty, x_n] \downarrow (-\infty, x].$$

Since μ is finite, continuity from above gives

$$\lim_{n\to\infty} F(x_n) = \lim_{n\to\infty} \mu((-\infty, x_n]) = \mu((-\infty, x]) = F(x).$$

Hence F is right-continuous. Finally,

$$(-\infty, b] = (-\infty, a] \,\dot\cup\, (a, b],$$

and finite additivity gives

$$\mu((a, b]) = \mu((-\infty, b]) - \mu((-\infty, a]) = F(b) - F(a).$$

□

Example 9 (Distribution function of a Dirac measure)**.** *Let $\mu = \delta_{x_0}$ be the Dirac measure at $x_0 \in \mathbb{R}$. Then*

$$F(x) = \delta_{x_0}((-\infty, x]) = \begin{cases} 0, & x < x_0, \\ 1, & x \ge x_0. \end{cases}$$

Thus an atom of a measure appears as a jump in its distribution function.

Remark 3 (Measures finite on bounded sets). A Borel measure need not be finite on all of $\mathbb{R}$. If μ is finite on every bounded Borel set, define the normalized distribution function

$$\widetilde{F}(x) = \begin{cases} \mu((0,x]), & x > 0, \\ 0, & x = 0, \\ -\mu((x,0]), & x < 0. \end{cases}$$

Then $\widetilde{F}$ is increasing and right-continuous, and

$$\mu((a,b]) = \widetilde{F}(b) - \widetilde{F}(a)$$

for all $a < b$. Adding a constant to $\widetilde{F}$ does not change these interval increments.

We now reverse the preceding construction at the level of the algebra of half-open intervals. Let $F\colon \mathbb{R} \to \mathbb{R}$ be increasing and right-continuous, and set

$$F(-\infty) = \lim_{x \to -\infty} F(x), \qquad F(\infty) = \lim_{x \to \infty} F(x),$$

where these limits are allowed to be infinite. Let $\mathcal{A}$ be the algebra of finite unions of intervals of the forms

$$(a,b], \qquad (-\infty, b], \qquad (a, \infty),$$

together with $\varnothing$. Equivalently, its members are finite unions of intervals $(a,b] \cap \mathbb{R}$ with $-\infty \le a < b \le \infty$.

Proposition 8 (The Lebesgue–Stieltjes premeasure)**.** *Suppose that*

$$E = \bigcup_{j=1}^{N} (a_j, b_j]$$

is a representation of $E \in \mathcal{A}$ as a finite union of pairwise disjoint half-open intervals, with extended endpoints allowed as above. Define

$$\mu_0(E) = \sum_{j=1}^{N} \big(F(b_j) - F(a_j)\big), \qquad \mu_0(\varnothing) = 0. \tag{3}$$

Then $\mu_0\colon \mathcal{A} \to [0,\infty]$ is well defined and is a premeasure.

Proof. We first show that the value in (3) is independent of the chosen representation. Suppose that a half-open interval is partitioned into finitely many pairwise disjoint half-open intervals:

$$(a, b] = \bigcup_{j=1}^{N} (a_j, b_j].$$

After relabeling, their endpoints satisfy

$$a = a_1 < b_1 = a_2 < b_2 = \cdots < b_{N-1} = a_N < b_N = b.$$

Hence the sum telescopes:

$$\sum_{j=1}^{N} \big(F(b_j) - F(a_j)\big) = F(b) - F(a).$$

The same argument applies when an endpoint is infinite, using the conventions for $F(-\infty)$ and $F(\infty)$.

Now suppose that

$$\bigcup_{i=1}^{N} I_i = \bigcup_{j=1}^{M} J_j,$$

where both families consist of pairwise disjoint half-open intervals. The nonempty intersections $I_i \cap J_j$ partition both sides. By the preceding telescoping identity,

$$\begin{aligned} \sum_{i=1}^{N} \mu_0(I_i) &= \sum_{i=1}^{N} \sum_{j=1}^{M} \mu_0(I_i \cap J_j) \\ &= \sum_{j=1}^{M} \sum_{i=1}^{N} \mu_0(I_i \cap J_j) = \sum_{j=1}^{M} \mu_0(J_j). \end{aligned}$$

Thus μ_0 is well defined; the same calculation also gives finite additivity.

It remains to prove countable additivity. After decomposing all sets involved into their interval components and intersecting with each component of the union, it is enough to consider pairwise disjoint half-open intervals I_j such that

$$(a, b] = \bigcup_{j=1}^{\infty} I_j.$$

Finite additivity and monotonicity give, for every n,

$$\mu_0((a,b]) \geq \mu_0\left(\bigcup_{j=1}^{n} I_j\right) = \sum_{j=1}^{n} \mu_0(I_j).$$

Letting $n \to \infty$ yields

$$\mu_0((a,b]) \geq \sum_{j=1}^{\infty} \mu_0(I_j). \tag{4}$$

For the reverse inequality, first suppose that $a, b \in \mathbb{R}$ and write $I_j = (a_j, b_j]$. Fix $\varepsilon > 0$. Right-continuity provides $0 < \delta < b - a$ and $\delta_j > 0$ such that

$$F(a+\delta) - F(a) < \varepsilon, \qquad F(b_j + \delta_j) - F(b_j) < \varepsilon 2^{-j}.$$

The open intervals $(a_j, b_j + \delta_j)$ cover the compact interval $[a+\delta, b]$, so finitely many of them suffice. After discarding redundant intervals, relabel this finite subcover from left to right with indices $1, \ldots, r$. Its total enlargement still satisfies

$$\sum_{j=1}^{r} \big(F(b_j + \delta_j) - F(b_j)\big) < \varepsilon,$$

and the ordering may be chosen so that

$$a_1 < a + \delta, \qquad a_{j+1} < b_j + \delta_j \ (1 \leq j < r), \qquad b < b_r + \delta_r.$$

Monotonicity of F then gives

$$\begin{aligned}
F(b) - F(a) &< F(b) - F(a+\delta) + \varepsilon \\
&\leq F(b_r + \delta_r) - F(a_1) + \varepsilon \\
&= F(b_r + \delta_r) - F(a_r) + \sum_{j=1}^{r-1} \big(F(a_{j+1}) - F(a_j)\big) + \varepsilon \\
&\leq \sum_{j=1}^{r} \big(F(b_j + \delta_j) - F(a_j)\big) + \varepsilon \\
&< \sum_{j=1}^{r} \big(F(b_j) - F(a_j)\big) + 2\varepsilon
\end{aligned}$$

$$\leq \sum_{j=1}^{\infty} \mu_0(I_j) + 2\varepsilon.$$

Letting $\varepsilon \downarrow 0$ proves the reverse of (4) for bounded intervals.

If the interval on the left is unbounded, apply the bounded case to its truncations. For example,

$$(-m, b] \uparrow (-\infty, b], \qquad \mu_0((-m, b]) = F(b) - F(-m) \uparrow F(b) - F(-\infty),$$

while the bounded case gives

$$\mu_0((-m, b]) = \sum_{j=1}^{\infty} \mu_0\big(I_j \cap (-m, b]\big) \leq \sum_{j=1}^{\infty} \mu_0(I_j).$$

Letting $m \to \infty$ proves the desired inequality for $(-\infty, b]$. The same argument uses $(a, m] \uparrow (a, \infty)$ or $(-m, m] \uparrow \mathbb{R}$ for the other unbounded cases. Therefore

$$\mu_0\left(\bigcup_{j=1}^{\infty} I_j\right) = \sum_{j=1}^{\infty} \mu_0(I_j),$$

whenever the union belongs to $\mathcal{A}$, so μ_0 is a premeasure. □

The premeasure extension theorem from the preceding lecture therefore applies to μ_0 on $\mathcal{A}$.

Remark 4 (Two Lebesgue–Stieltjes examples). The Borel measure determined by the increments $F(b) - F(a)$ is denoted by μ_F, and its completion is called the *Lebesgue–Stieltjes measure* associated with F.

(i) For $F(x) = x$,

$$\mu_F((a, b]) = b - a.$$

The completed measure is Lebesgue measure on $\mathbb{R}$.

(ii) For the right-continuous step function

$$F(x) = \mathbf{1}_{[x_0, \infty)}(x),$$

the associated measure is the Dirac measure $\mu_F = \delta_{x_0}$.

6. September 10, 2026

We finish the construction begun in the preceding lecture. Recall that an increasing, right-continuous function $F\colon \mathbb{R} \to \mathbb{R}$ defines a premeasure on the algebra of finite unions of half-open intervals by

$$\mu_0((a,b]) = F(b) - F(a).$$

The extension theorem and its uniqueness statement give the following result.

Theorem 6 (Lebesgue–Stieltjes measure)**.** *Let $F\colon \mathbb{R} \to \mathbb{R}$ be increasing and right-continuous. There is a unique Borel measure μ_F on $\mathbb{R}$, finite on bounded sets, such that*

$$\mu_F((a,b]) = F(b) - F(a)$$

for every $a < b$. Its completion $\overline{\mu}_F$ is called the Lebesgue–Stieltjes measure *associated with F.*

Proof. Proposition 8 shows that the interval increments of F define a premeasure on the algebra $\mathcal{A}$ of finite unions of half-open intervals. The premeasure extension theorem therefore produces a measure on

$$\mathcal{M}(\mathcal{A}) = \mathcal{B}_{\mathbb{R}}.$$

It is finite on bounded intervals because F is real-valued. In particular, the premeasure is σ-finite, so uniqueness follows from Theorem 5. $\square$

The measure depends only on the increments of F. Thus

$$\mu_F = \mu_G \quad \Longleftrightarrow \quad F - G \text{ is constant.}$$

Conversely, every Borel measure μ on $\mathbb{R}$ that is finite on bounded sets arises in this way. One convenient normalization is

$$F(x) = \begin{cases} \mu((0,x]), & x > 0, \\ 0, & x = 0, \\ -\mu((x,0]), & x < 0. \end{cases}$$

Then F is increasing and right-continuous, and uniqueness of the extension gives $\mu = \mu_F$ on $\mathcal{B}_{\mathbb{R}}$.

One could instead work with an increasing, left-continuous function G. With that convention the natural intervals are left-closed and right-open, and

$$\mu_G([a,b)) = G(b) - G(a).$$

Example 10. *If $F(x) = x$, then*

$$\mu_F((a,b]) = b - a.$$

The completion of μ_F is Lebesgue measure.

Example 11. *If $F(x) = \mathbf{1}_{[x_0,\infty)}(x)$, then $\mu_F = \delta_{x_0}$.*

6.1. **Regularity of Lebesgue–Stieltjes measures.** Fix an increasing, right-continuous F, and write μ for the completed measure $\overline{\mu}_F$. Its domain will be denoted by $\mathcal{M}_F$. The outer-measure construction gives, for every $E \in \mathcal{M}_F$,

$$\mu(E) = \inf\left\{\sum_{j=1}^{\infty}\big(F(b_j) - F(a_j)\big) : E \subseteq \bigcup_{j=1}^{\infty}(a_j, b_j]\right\}. \tag{5}$$

There is an equivalent formula involving open intervals.

Lemma 2 (Open covering formula)**.** *For every $E \in \mathcal{M}_F$,*

$$\mu(E) = \inf\left\{\sum_{j=1}^{\infty}\mu((a_j,b_j)) : E \subseteq \bigcup_{j=1}^{\infty}(a_j,b_j)\right\}.$$

Proof. Let $V(E)$ denote the infimum on the right. If $E \subseteq \bigcup_j (a_j, b_j)$, then countable subadditivity gives

$$\mu(E) \le \mu\left(\bigcup_{j=1}^{\infty}(a_j,b_j)\right) \le \sum_{j=1}^{\infty}\mu((a_j,b_j)).$$

Taking the infimum shows that $\mu(E) \le V(E)$.

For the reverse inequality, suppose first that $\mu(E) < \infty$, and fix $\varepsilon > 0$. By (5), there are intervals $(a_j, b_j]$ covering E such that

$$\sum_{j=1}^{\infty}\mu((a_j,b_j]) < \mu(E) + \varepsilon.$$

Right-continuity of F allows us to choose $\delta_j > 0$ with

$$F(b_j + \delta_j) - F(b_j) < \varepsilon 2^{-j}.$$

The open intervals $(a_j, b_j + \delta_j)$ still cover E, and

$$\begin{aligned}\sum_{j=1}^{\infty} \mu((a_j, b_j + \delta_j)) &\leq \sum_{j=1}^{\infty} \big(F(b_j + \delta_j) - F(a_j)\big) \\ &< \sum_{j=1}^{\infty} \mu((a_j, b_j]) + \varepsilon \\ &< \mu(E) + 2\varepsilon.\end{aligned}$$

Hence $V(E) \leq \mu(E)$. If $\mu(E) = \infty$, the already proved inequality $\mu(E) \leq V(E)$ forces $V(E) = \infty$. □

Theorem 7 (Regularity)**.** *Let $E \in \mathcal{M}_F$.*

(i) Outer regularity:

$$\mu(E) = \inf\{\mu(U) : U \supseteq E,\ U \text{ open}\}.$$

(ii) Inner regularity:

$$\mu(E) = \sup\{\mu(K) : K \subseteq E,\ K \text{ compact}\}.$$

Proof. If $E \subseteq U$, then $\mu(E) \leq \mu(U)$. Conversely, the open covering formula provides open intervals (a_j, b_j) such that $E \subseteq U = \bigcup_j (a_j, b_j)$ and

$$\mu(U) \leq \sum_{j=1}^{\infty} \mu((a_j, b_j)) < \mu(E) + \varepsilon$$

whenever $\mu(E) < \infty$. The infinite-measure case is immediate. This proves outer regularity.

We first prove inner regularity when E is bounded. Then $\overline{E}$ has finite measure. By outer regularity, there is an open set U containing $\overline{E} \setminus E$ such that

$$\mu(U) < \mu(\overline{E} \setminus E) + \varepsilon.$$

The set

$$K = \overline{E} \setminus U$$

is closed and bounded, hence compact, and $K \subseteq E$. Since all relevant measures are finite,

$$\begin{aligned}\mu(K) &= \mu(\overline{E}) - \mu(\overline{E} \cap U) \\ &\geq \mu(\overline{E}) - \mu(U) \\ &> \mu(\overline{E}) - \mu(\overline{E} \setminus E) - \varepsilon = \mu(E) - \varepsilon.\end{aligned}$$

For arbitrary E, apply the bounded case to $E \cap [-n, n]$. These sets increase to E, so continuity from below gives

$$\mu(E) = \lim_{n\to\infty} \mu(E \cap [-n, n]).$$

Compact subsets approximating the bounded truncations therefore give the desired supremum, whether $\mu(E)$ is finite or infinite. □

Regularity gives useful Borel representatives for every set in the completion.

Corollary 2. *If $E \in \mathcal{M}_F$, then there are a G_δ set G, an F_σ set H, and null sets N_1, N_2 such that*

$$H \subseteq E \subseteq G, \qquad E = G \setminus N_1 = H \cup N_2.$$

In particular, every Lebesgue–Stieltjes measurable set differs from a Borel set by a null set.

Proof. Decompose E into the bounded pieces $E_k = E \cap [k, k+1)$, $k \in \mathbb{Z}$. For every n and k, outer regularity gives an open set $U_{n,k} \supseteq E_k$ for which

$$\mu(U_{n,k} \setminus E_k) < 2^{-n-|k|-2}.$$

Set

$$U_n = \bigcup_{k\in\mathbb{Z}} U_{n,k}, \qquad G = \bigcap_{n=1}^{\infty} U_n.$$

Then G is a G_δ set containing E, and $\mu(G \setminus E) = 0$. Thus $E = G \setminus N_1$, where $N_1 = G \setminus E$ is null. Apply the same argument to E^c and take complements to find an F_σ set $H \subseteq E$ such that $E \setminus H$ is null. Taking $N_2 = E \setminus H$ completes the proof. □

6.2. **Lebesgue measure.** Let m denote Lebesgue measure on the Lebesgue σ-algebra $\mathcal{L}$. Thus m is the completion of the Borel measure generated by $F(x) = x$.

Theorem 8 (Translation and dilation invariance)**.** *If $E \in \mathcal{L}$ and $s \in \mathbb{R}$, then $E + s \in \mathcal{L}$ and*

$$m(E + s) = m(E).$$

If $\lambda \in \mathbb{R} \setminus \{0\}$, then $\lambda E \in \mathcal{L}$ and

$$m(\lambda E) = |\lambda| m(E).$$

For $\lambda = 0$, the set λE is either empty or a singleton and therefore has measure zero. Thus the dilation formula also holds for $\lambda = 0$ when $m(E) < \infty$ (or with the convention $0 \cdot \infty = 0$).

Proof. Translation by s is a homeomorphism, so it preserves Borel sets. On $\mathcal{B}_{\mathbb{R}}$ define

$$m_s(E) = m(E + s).$$

This is a Borel measure, and for every half-open interval,

$$m_s((a, b]) = m((a + s, b + s]) = b - a = m((a, b]).$$

Uniqueness of the premeasure extension gives $m_s = m$ on $\mathcal{B}_{\mathbb{R}}$.

If N is a Borel null set, then $N + s$ is again a Borel null set. Every $E \in \mathcal{L}$ has the form

$$E = B \cup N,$$

where B is Borel and N is contained in a Borel null set. It follows that $E + s$ is Lebesgue measurable and $m(E + s) = m(B + s) = m(B) = m(E)$.

Now fix $\lambda \neq 0$. The map $x \mapsto \lambda x$ is also a homeomorphism. Singletons have measure zero because

$$m(\{x\}) \leq m((x - \varepsilon, x]) = \varepsilon$$

for every $\varepsilon > 0$. Consequently every interval with endpoints $c < d$, regardless of the choice of open or closed endpoints, has measure $d - c$.

Hence

$$m_\lambda(E) = \frac{1}{|\lambda|} m(\lambda E), \qquad E \in \mathcal{B}_{\mathbb{R}},$$

is a Borel measure satisfying $m_\lambda((a,b]) = b - a$. Uniqueness gives $m_\lambda = m$ on Borel sets. As above, dilation carries Borel null sets to Borel null sets, so the identity extends to every $E \in \mathcal{L}$. □

Example 12 (Countable sets)**.** *Every singleton is null, and therefore every countable set is null. In particular,*

$$m(\mathbb{Q}) = 0.$$

More concretely, if $\mathbb{Q} \cap (0,1) = \{q_1, q_2, \ldots\}$ *and* $\varepsilon > 0$*, then*

$$U_\varepsilon = \bigcup_{j=1}^{\infty} (q_j - \varepsilon 2^{-j}, q_j + \varepsilon 2^{-j})$$

is an open dense set containing $\mathbb{Q} \cap (0,1)$*, while*

$$m(U_\varepsilon) \le \sum_{j=1}^{\infty} 2\varepsilon 2^{-j} = 2\varepsilon.$$

Thus an open dense set can have arbitrarily small measure.

7. September 15, 2026

We begin with several examples illustrating the difference between measure-theoretic size and topological size. Throughout, m denotes Lebesgue measure and $\mathcal{L}$ denotes the Lebesgue σ-algebra.

7.1. Measure and Baire category.

Enumerate the rationals as $\mathbb{Q} = \{q_1, q_2, \ldots\}$. For $n \in \mathbb{N}$, define

$$U_n = \bigcup_{k=1}^{\infty} \left(q_k - 2^{-n-k-2}, q_k + 2^{-n-k-2}\right).$$

Each U_n is open and dense because it contains $\mathbb{Q}$, but

$$m(U_n) \le \sum_{k=1}^{\infty} 2^{-n-k-1} = 2^{-n-1}.$$

The set

$$U = \bigcap_{n=1}^{\infty} U_n$$

is a dense G_δ set by the Baire category theorem, yet $m(U) = 0$. It is necessarily uncountable: a countable subset of $\mathbb{R}$ is meager, whereas a dense G_δ subset of $\mathbb{R}$ is comeager.

This example has a complementary form on $[0, 1]$. Let

$$F_n = [0, 1] \setminus U_n, \qquad F = \bigcup_{n=1}^{\infty} F_n = [0, 1] \setminus U.$$

Because U_n is open and dense, F_n is closed and nowhere dense. Thus F is meager, being a countable union of nowhere dense sets, but

$$m(F) = 1.$$

Thus the topological and measure-theoretic notions of size need not agree.

7.2. **The Cantor set.** The middle-thirds Cantor set $C \subseteq [0, 1]$ can be described in two equivalent ways. First, set $C_0 = [0, 1]$ and obtain C_n from C_{n-1} by deleting the open middle third of each interval component. Then

$$C = \bigcap_{n=0}^{\infty} C_n.$$

Equivalently,

$$C = \left\{ \sum_{j=1}^{\infty} a_j 3^{-j} : a_j \in \{0, 2\} \right\}.$$

Proposition 9. *The Cantor set has the following properties.*

(a) $m(C) = 0$.
(b) C *is compact, has empty interior, and is totally disconnected.*
(c) $|C| = |\mathbb{R}|$.

Proof. The set C_n consists of 2^n closed intervals, each of length 3^{-n}. By continuity from above,

$$m(C) = \lim_{n\to\infty} m(C_n) = \lim_{n\to\infty} \left(\frac{2}{3}\right)^n = 0.$$

Each C_n is compact, so their nested intersection C is compact. Since every nonempty interval has positive measure, the equality $m(C) = 0$ implies that C has empty interior. Every connected subset of $\mathbb{R}$ is an interval; since C contains no nontrivial interval, each connected component of C is a singleton. Thus C is totally disconnected.

Finally, the map

$$(b_j)_{j=1}^{\infty} \longmapsto \sum_{j=1}^{\infty} 2b_j 3^{-j}, \qquad b_j \in \{0,1\},$$

is a bijection from $\{0,1\}^{\mathbb{N}}$ onto C. Since the set of binary sequences has the cardinality of $\mathbb{R}$, so does C. □

The Cantor set supports a function that is continuous and increasing but whose derivative vanishes almost everywhere.

Definition 11 (Cantor function)**.** *If $x \in C$ has ternary expansion*

$$x = \sum_{j=1}^{\infty} 2b_j 3^{-j}, \qquad b_j \in \{0,1\},$$

define

$$\Phi(x) = \sum_{j=1}^{\infty} b_j 2^{-j}.$$

Every component of $[0,1] \setminus C$ is an open interval whose endpoints lie in C and have the same value under the preceding formula. Extend Φ to be constant on each such interval. The resulting map $\Phi\colon [0,1] \to [0,1]$ is the Cantor function*.*

The Cantor function is continuous and increasing, with $\Phi(0) = 0$ and $\Phi(1) = 1$. On the open set $[0,1] \setminus C$ it is locally constant. Since $m(C) = 0$, it follows that

$$\Phi'(x) = 0 \quad \text{for almost every } x \in [0,1].$$

Nevertheless,

$$\Phi(1) - \Phi(0) = 1 \neq 0 = \int_0^1 \Phi'(x)\, dx.$$

Thus continuity alone is not sufficient for the strongest form of the fundamental theorem of calculus.

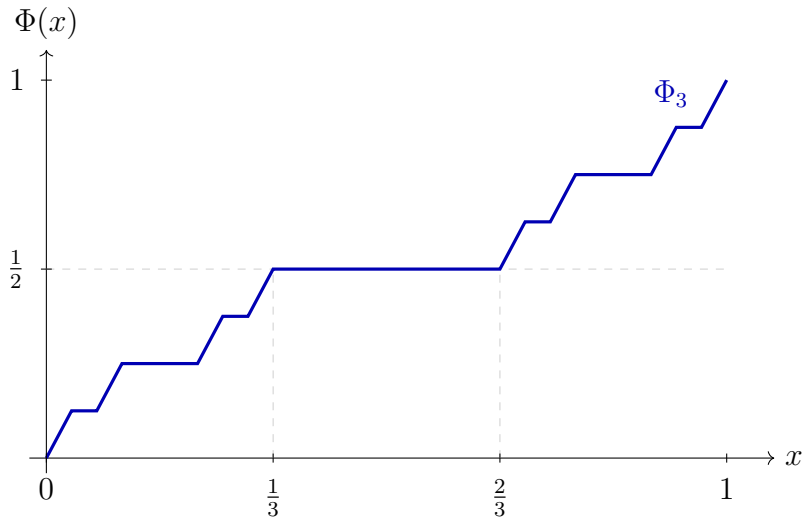


FIGURE 1. The third-stage polygonal approximation Φ_3 to the Cantor function. At every subsequent stage, each sloping segment is replaced by a smaller copy of the first-stage pattern; the limit is constant on every component of $[0,1] \setminus C$.

Example 13 (A fat Cantor set)**.** *Set $K_0 = [0,1]$. At stage $n \geq 1$, remove from the center of each interval component of K_{n-1} an open interval of length 4^{-n}, and call the remaining set K_n. Thus the first step removes the middle interval $(3/8, 5/8)$ and leaves*

$$K_1 = \left[0, \frac{3}{8}\right] \cup \left[\frac{5}{8}, 1\right].$$

There are 2^{n-1} intervals removed at stage n, each of length 4^{-n}. Hence, for the limit set $K = \bigcap_{n=0}^{\infty} K_n$,

$$m(K) = 1 - \sum_{n=1}^{\infty} 2^{n-1} 4^{-n} = 1 - \sum_{n=1}^{\infty} 2^{-n-1} = \frac{1}{2}.$$

The lengths of the interval components tend to zero, while every component splits into two components at the next stage. Consequently, K is compact, perfect, has empty interior, and is totally disconnected. Thus

K has all the topological features of the usual Cantor set but has positive measure. It is called the fat Cantor set *(or the* Smith–Volterra–Cantor set*).*

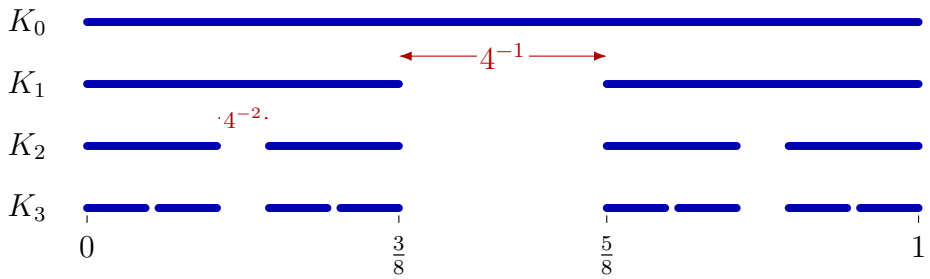


FIGURE 2. The first stages of the fat Cantor construction. The deleted intervals become shorter quickly enough that the limiting set retains measure 1/2.

7.3. **The idea of Lebesgue integration.** Let $(X, \mathcal{M}, \mu)$ be a measure space. For $E \in \mathcal{M}$, the *characteristic function* of E is

$$\mathbf{1}_E(x) = \begin{cases} 1, & x \in E, \\ 0, & x \notin E. \end{cases}$$

The integral is required to satisfy

$$\int_X \mathbf{1}_E \, d\mu = \mu(E).$$

This suggests how to integrate a more general function. Rather than partitioning the domain into short intervals, as in Riemann integration, Lebesgue integration partitions the range. For a mesh size $\varepsilon > 0$ and $k \in \mathbb{Z}$, consider the level sets

$$E_k = \{x \in X : k\varepsilon \le f(x) < (k+1)\varepsilon\}.$$

When these sets belong to $\mathcal{M}$, the function can be approximated by a simple function, and formally

$$\int_X f \, d\mu \approx \sum_{k \in \mathbb{Z}} k\varepsilon \, \mu(E_k).$$

The requirement that the level sets be measurable leads to the notion of a measurable function.

7.4. **Measurable functions.**

Definition 12. *Let $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be measurable spaces. A map $f\colon X \to Y$ is* $(\mathcal{M}, \mathcal{N})$-measurable *if*

$$f^{-1}(E) \in \mathcal{M}$$

for every $E \in \mathcal{N}$.

For a real- or complex-valued function, the codomain is equipped with its Borel σ-algebra. Thus $f\colon X \to \mathbb{R}$ is $\mathcal{M}$-measurable when it is $(\mathcal{M}, \mathcal{B}_{\mathbb{R}})$-measurable, and similarly for $f\colon X \to \mathbb{C}$.

The extended real line

$$\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$$

is equipped with its Borel σ-algebra. Its topology may be described by the metric

$$d(x, y) = |\arctan x - \arctan y|, \qquad \arctan(\pm\infty) = \pm\frac{\pi}{2}.$$

For functions on $\mathbb{R}$, the terminology is worth distinguishing:

- f is *Borel measurable* if it is $(\mathcal{B}_{\mathbb{R}}, \mathcal{B}_{\mathbb{R}})$-measurable;
- f is *Lebesgue measurable* if it is $(\mathcal{L}, \mathcal{B}_{\mathbb{R}})$-measurable.

The codomain still carries the Borel σ-algebra in the second definition. In particular, although a composition of appropriately measurable maps is measurable, the composition of two Lebesgue measurable functions $\mathbb{R} \to \mathbb{R}$ need not be Lebesgue measurable: the inverse image under the outer function is only known to be a Lebesgue set, while the inner function is only required to pull back Borel sets.

Lemma 3 (Checking a generating family)**.** *Suppose $\mathcal{N} = \mathcal{M}(\mathcal{E})$ for some $\mathcal{E} \subseteq 2^Y$. A map $f\colon X \to Y$ is $(\mathcal{M}, \mathcal{N})$-measurable if and only if*

$$f^{-1}(E) \in \mathcal{M}$$

for every $E \in \mathcal{E}$.

Proof. Only the reverse implication requires proof. Set

$$\mathcal{A} = \{E \subseteq Y : f^{-1}(E) \in \mathcal{M}\}.$$

Since preimages preserve complements and countable unions, $\mathcal{A}$ is a σ-algebra. By hypothesis it contains $\mathcal{E}$, so it contains $\mathcal{M}(\mathcal{E}) = \mathcal{N}$. □

Corollary 3. *If X and Y are topological spaces and $f\colon X \to Y$ is continuous, then f is $(\mathcal{B}_X, \mathcal{B}_Y)$-measurable.*

Proof. The Borel σ-algebra $\mathcal{B}_Y$ is generated by the open subsets of Y, and the inverse image of an open set under a continuous map is open. □

Proposition 10 (Half-line criterion)**.** *Let $f\colon X \to \overline{\mathbb{R}}$. The following are equivalent.*

(i) f *is measurable.*
(ii) $\{x : f(x) > a\} \in \mathcal{M}$ *for every* $a \in \mathbb{R}$.
(iii) $\{x : f(x) \geq a\} \in \mathcal{M}$ *for every* $a \in \mathbb{R}$.
(iv) $\{x : f(x) < a\} \in \mathcal{M}$ *for every* $a \in \mathbb{R}$.
(v) $\{x : f(x) \leq a\} \in \mathcal{M}$ *for every* $a \in \mathbb{R}$.

Proof. Each of the four corresponding families of half-lines generates the Borel σ-algebra of $\overline{\mathbb{R}}$. The conclusion follows from Lemma 3. □

Finally, if $(X_i, \mathcal{M}_i)_{i\in I}$ are measurable spaces, the product σ-algebra $\bigotimes_{i\in I} \mathcal{M}_i$ is precisely the smallest σ-algebra on $\prod_i X_i$ for which every coordinate projection

$$\pi_j\colon \prod_{i\in I} X_i \longrightarrow X_j$$

is measurable. This is the measurable-space analogue of the fact that the product topology is the smallest topology making all coordinate projections continuous.

8. September 17, 2026

We continue the study of measurable functions and then introduce the building blocks of the Lebesgue integral. Throughout, $(X, \mathcal{M})$ is a measurable space. A real- or complex-valued function is understood to have the Borel σ-algebra on its codomain.

Recall that the Borel σ-algebra on $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$ is generated by either of the families

$$\{(a, +\infty] : a \in \mathbb{R}\} \quad \text{and} \quad \{[-\infty, a] : a \in \mathbb{R}\}.$$

Thus the half-line criterion from the preceding lecture applies equally well to extended-real-valued functions.

8.1. **Operations on measurable functions.** Let $(Y_i, \mathcal{N}_i)_{i \in I}$ be measurable spaces, and give

$$Y = \prod_{i \in I} Y_i$$

the product σ-algebra $\mathcal{N} = \bigotimes_{i \in I} \mathcal{N}_i$. If $\pi_i \colon Y \to Y_i$ is the ith coordinate projection, then $\mathcal{N}$ is generated by the cylinder sets

$$\pi_i^{-1}(E), \qquad i \in I, \quad E \in \mathcal{N}_i.$$

Lemma 4 (Coordinate criterion for product measurability)**.** *A map $f \colon X \to Y$ is $(\mathcal{M}, \bigotimes_{i \in I} \mathcal{N}_i)$-measurable if and only if every coordinate map*

$$\pi_i \circ f \colon X \longrightarrow Y_i$$

is $(\mathcal{M}, \mathcal{N}_i)$-measurable.

Proof. If f is measurable, then $\pi_i \circ f$ is measurable because each projection π_i is measurable. Conversely, suppose all the coordinate maps are measurable. For every generating cylinder,

$$f^{-1}\big(\pi_i^{-1}(E)\big) = (\pi_i \circ f)^{-1}(E) \in \mathcal{M}.$$

Lemma 3 now shows that f is measurable. □

Corollary 4. *A function $f \colon X \to \mathbb{C}$ is measurable if and only if* $\operatorname{Re} f$ *and* $\operatorname{Im} f$ *are measurable.*

Proof. Under the identification $\mathbb{C} \cong \mathbb{R}^2$,

$$\mathcal{B}_{\mathbb{C}} = \mathcal{B}_{\mathbb{R}} \otimes \mathcal{B}_{\mathbb{R}}.$$

The claim is therefore the two-coordinate case of Lemma 4. □

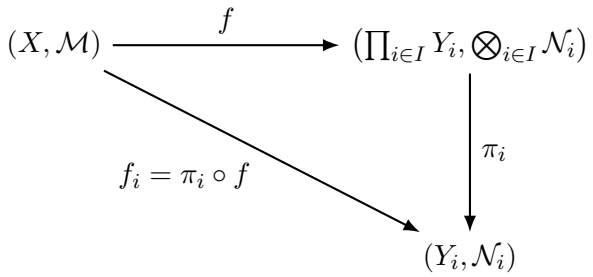


FIGURE 3. Measurability into a product can be checked one coordinate at a time.

Proposition 11 (Algebra and lattice operations)**.** *If $f, g\colon X \to \mathbb{R}$ are measurable, then so are*

$$f+g, \quad fg, \quad f^+ = \max\{f, 0\}, \quad f^- = \max\{-f, 0\}, \quad |f|,$$

and $\operatorname{sgn}(f)$*.*

Proof. By Lemma 4, the map

$$F = (f, g)\colon X \longrightarrow \mathbb{R}^2$$

is measurable. Addition and multiplication are continuous maps from $\mathbb{R}^2$ to $\mathbb{R}$, so they are Borel measurable. Hence

$$f + g = \Sigma \circ F, \qquad fg = \Pi \circ F,$$

where $\Sigma(s,t) = s+t$ and $\Pi(s,t) = st$, are measurable. The remaining functions follow by composition with the Borel maps

$$t \longmapsto \max\{t, 0\}, \quad t \longmapsto \max\{-t, 0\}, \quad t \longmapsto |t|,$$

and with the Borel measurable sign function. Notice also that $f = f^+ - f^-$ and $|f| = f^+ + f^-$. □

The same argument, together with Corollary 4, shows that sums and products of complex-valued measurable functions are measurable.

Proposition 12 (Countable suprema and limits)**.** *Let $f_n\colon X \to \overline{\mathbb{R}}$ be measurable for every $n \in \mathbb{N}$. Then the functions*

$$\sup_n f_n, \qquad \inf_n f_n, \qquad \limsup_{n\to\infty} f_n, \qquad \liminf_{n\to\infty} f_n$$

are measurable. In particular, if $f_n \to f$ pointwise, then f is measurable.

Proof. For $a \in \mathbb{R}$,

$$\left\{\sup_n f_n > a\right\} = \bigcup_{n=1}^{\infty} \{f_n > a\}, \qquad \left\{\inf_n f_n < a\right\} = \bigcup_{n=1}^{\infty} \{f_n < a\}.$$

The half-line criterion implies that the pointwise supremum and infimum are measurable. Now use the identities

$$\limsup_{n\to\infty} f_n = \inf_{N\geq 1} \sup_{n\geq N} f_n, \qquad \liminf_{n\to\infty} f_n = \sup_{N\geq 1} \inf_{n\geq N} f_n.$$

If $f_n \to f$ pointwise, both expressions equal f. □

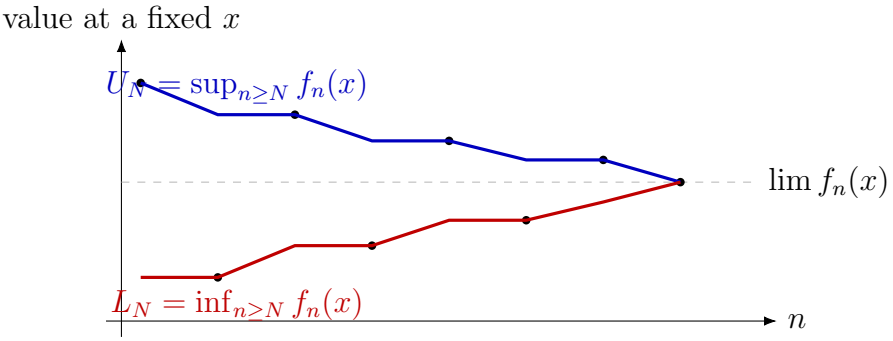


FIGURE 4. The tail supremum decreases and the tail infimum increases. Their limits are the pointwise lim sup and lim inf; the drawing shows the convergent case.

8.2. **Simple functions.** For $E \in \mathcal{M}$, continue to write $\mathbf{1}_E$ for the characteristic function of E.

Definition 13. *A measurable function $\phi\colon X \to \mathbb{C}$ is* simple *if it has finite range. Equivalently, it can be written as a finite linear combination*

$$\phi = \sum_{j=1}^{N} a_j \mathbf{1}_{E_j}, \qquad a_j \in \mathbb{C}, \quad E_j \in \mathcal{M}.$$

If $\{a_1, \dots, a_N\}$ is the set of distinct values of ϕ and $E_j = \phi^{-1}(\{a_j\})$, then the sets E_j are pairwise disjoint and cover X. This is the standard representation *of ϕ.*

Sums, products, absolute values, maxima, and minima of simple functions are again simple. The standard representation makes a simple function look like a finite collection of horizontal levels over a measurable partition of its domain.

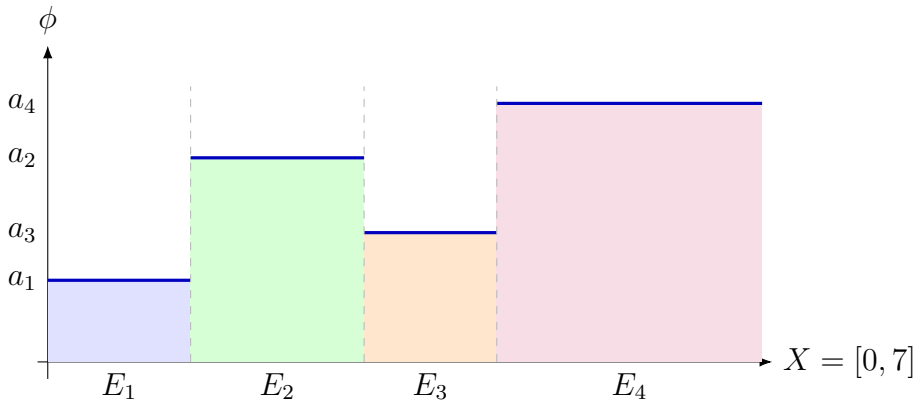


FIGURE 5. A simple function $\phi = \sum_{j=1}^{4} a_j \mathbf{1}_{E_j}$ in standard form. The sets E_j need not be intervals in a general measurable space.

Simple functions are sufficient to approximate every measurable function.

Theorem 9 (Approximation by simple functions)**.** *(i) If $f\colon X \to [0, +\infty]$ is measurable, then there are nonnegative simple functions ϕ_n such that*

$$0 \le \phi_1 \le \phi_2 \le \cdots \le f, \qquad \phi_n(x) \longrightarrow f(x)$$

for every $x \in X$. The convergence is uniform on every set on which f is bounded.

(ii) If $f\colon X \to \mathbb{C}$ is measurable, then there are simple functions $\phi_n\colon X \to \mathbb{C}$ such that

$$0 \le |\phi_1| \le |\phi_2| \le \cdots \le |f|, \qquad \phi_n(x) \longrightarrow f(x)$$

for every $x \in X$. Again, convergence is uniform on every set on which $|f|$ is bounded.

Proof. For (i), divide the interval $[0, n]$ into dyadic pieces of length 2^{-n}. Set

$$E_{n,k} = f^{-1}\big((k2^{-n}, (k+1)2^{-n}]\big), \qquad 0 \le k < n2^n,$$

and $F_n = f^{-1}((n, +\infty])$. Define

$$\phi_n = \sum_{k=0}^{n2^n-1} k2^{-n}\mathbf{1}_{E_{n,k}} + n\mathbf{1}_{F_n}. \tag{6}$$

Each ϕ_n is a nonnegative simple function. The dyadic grids refine one another and the truncation height increases, so $\phi_n \le \phi_{n+1} \le f$. Moreover,

$$0 \le f(x) - \phi_n(x) \le 2^{-n} \quad \text{whenever } f(x) \le n.$$

It follows that $\phi_n \uparrow f$ pointwise. If $f \le M$ on a set E, then for $n \ge M$,

$$\sup_{x\in E} |f(x) - \phi_n(x)| \le 2^{-n},$$

which proves the asserted uniform convergence.

For (ii), write

$$f = (\operatorname{Re} f)^+ - (\operatorname{Re} f)^- + i(\operatorname{Im} f)^+ - i(\operatorname{Im} f)^-.$$

Apply (i) to these four nonnegative measurable functions to obtain increasing simple approximants u_n, v_n, w_n, z_n, respectively, and set

$$\phi_n = (u_n - v_n) + i(w_n - z_n).$$

At each point, at most one of u_n, v_n is nonzero and at most one of w_n, z_n is nonzero. Consequently $|\phi_n|$ increases with n and is bounded by $|f|$. The convergence statements follow from those for the four real components. $\square$

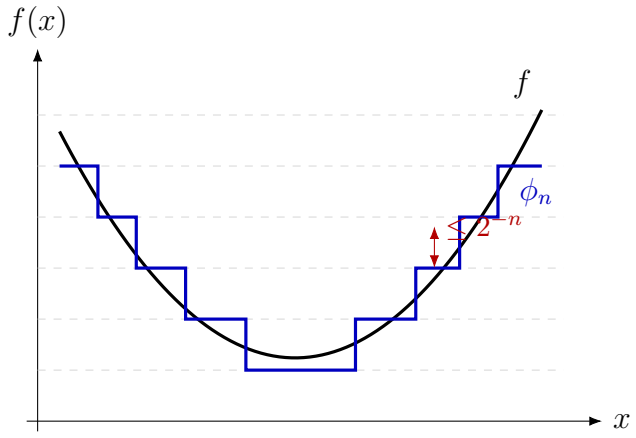


FIGURE 6. The approximation in (6) rounds f down to a dyadic level and truncates it at height n. Below the truncation height, the vertical error is at most 2^{-n}.

8.3. **Nonnegative integration.** Now fix a measure space $(X, \mathcal{M}, \mu)$ and write

$$L^+ = \{f \colon X \to [0, +\infty] : f \text{ is measurable}\}.$$

Definition 14 (Integral of a nonnegative simple function)**.** *Let $\phi \in L^+$ be simple, with standard representation*

$$\phi = \sum_{j=1}^{N} a_j \mathbf{1}_{E_j}.$$

Define

$$\int_X \phi \, d\mu = \sum_{j=1}^{N} a_j \mu(E_j),$$

using the convention $0 \cdot \infty = 0$. For $A \in \mathcal{M}$, set

$$\int_A \phi \, d\mu = \int_X \phi \mathbf{1}_A \, d\mu.$$

The standard representation is unique up to the ordering of its nonempty level sets, so this definition is unambiguous. Geometrically, each term $a_j \mu(E_j)$ is the measure-theoretic analogue of the area of a rectangle.

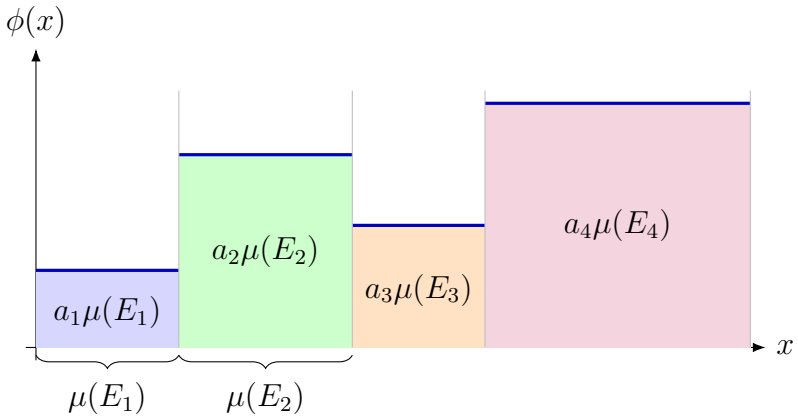


FIGURE 7. For Lebesgue measure on an interval, the simple integral is literally the sum of the shaded rectangular areas. The definition $\sum_j a_j\mu(E_j)$ makes sense on every measure space.

Definition 15 (Integral of a nonnegative measurable function)**.** *For $f \in L^+$, define*

$$\int_X f\,d\mu = \sup\left\{\int_X \phi\,d\mu : \phi \text{ is simple and } 0 \le \phi \le f\right\}.$$

This value belongs to $[0, +\infty]$. For $A \in \mathcal{M}$, define

$$\int_A f\,d\mu = \int_X f\mathbf{1}_A\,d\mu.$$

Theorem 9 explains why the supremum uses precisely the right class of building blocks: every $f \in L^+$ is the pointwise increasing limit of simple functions below it.

Example 14 (Three measures and their integrals)**.** *(i) If $\mu = \delta_{x_0}$ is the Dirac measure at $x_0 \in X$, then*

$$\int_X f\,d\delta_{x_0} = f(x_0) \qquad (f \in L^+).$$

Thus a Dirac integral samples the function at one point.

(ii) If $\mu = m$ is Lebesgue measure on $\mathbb{R}$, then the integral of a nonnegative simple function is the usual sum of rectangular areas, as in Figure 7.

(iii) Since $\mathbb{Q}$ is null but the fat Cantor set K from Example 13 has measure $1/2$,

$$\int_{\mathbb{R}} \mathbf{1}_{\mathbb{Q}}\, dm = 0, \qquad \int_{\mathbb{R}} \mathbf{1}_{K}\, dm = \frac{1}{2}.$$

Figure 2 shows the construction of K.

We finish with properties that follow directly from the finite sum defining the integral of a simple function.

Proposition 13 (Elementary properties of the simple integral)**.** *Let $\phi, \psi \in L^+$ be simple and let $c \geq 0$.*

(a) $\displaystyle\int_X c\phi\, d\mu = c \int_X \phi\, d\mu.$

(b) $\displaystyle\int_X (\phi + \psi)\, d\mu = \int_X \phi\, d\mu + \int_X \psi\, d\mu.$

(c) If $\phi \leq \psi$, then $\displaystyle\int_X \phi\, d\mu \leq \int_X \psi\, d\mu.$

(d) The set function

$$\nu(A) = \int_A \phi\, d\mu, \qquad A \in \mathcal{M},$$

is a measure on $(X, \mathcal{M})$.

Proof. Part (a) is immediate. For (b), write the standard representations as

$$\phi = \sum_{j=1}^{N} a_j \mathbf{1}_{E_j}, \qquad \psi = \sum_{k=1}^{M} b_k \mathbf{1}_{F_k}.$$

The sets $E_j \cap F_k$ give a common measurable partition, and

$$\phi + \psi = \sum_{j,k} (a_j + b_k) \mathbf{1}_{E_j \cap F_k}.$$

Finite additivity of μ gives

$$\begin{aligned} \int_X (\phi + \psi)\, d\mu &= \sum_{j,k} (a_j + b_k) \mu(E_j \cap F_k) \\ &= \sum_{j} a_j \mu(E_j) + \sum_{k} b_k \mu(F_k) \\ &= \int_X \phi\, d\mu + \int_X \psi\, d\mu. \end{aligned}$$

The same common partition proves (c): whenever $E_j \cap F_k \neq \varnothing$, the inequality $\phi \leq \psi$ implies $a_j \leq b_k$, so comparison term by term gives the result.

Finally,

$$\nu(A) = \int_X \phi \mathbf{1}_A \, d\mu = \sum_{j=1}^{N} a_j \mu(E_j \cap A).$$

Each map $A \mapsto \mu(E_j \cap A)$ is a measure. Therefore their finite nonnegative linear combination ν is a measure, proving (d). □

9. September 22, 2026

Throughout, $(X, \mathcal{M}, \mu)$ is a measure space, and

$$L^+ = \{f \colon X \to [0, +\infty] : f \text{ is measurable}\}.$$

We continue to use $\mathbf{1}_E$ for the characteristic function of a measurable set E and the convention $0 \cdot \infty = 0$.

9.1. **Nonnegative integrals and infinite values.**

Recall that, for a nonnegative simple function in standard form,

$$\phi = \sum_{j=1}^{N} a_j \mathbf{1}_{E_j}, \qquad \int_X \phi \, d\mu = \sum_{j=1}^{N} a_j \mu(E_j).$$

Moreover, $A \mapsto \int_A \phi \, d\mu$ is a measure. For $f \in L^+$,

$$\int_X f \, d\mu = \sup \left\{ \int_X \phi \, d\mu : \phi \text{ is simple, } 0 \leq \phi \leq f \right\}.$$

In particular, $f \leq g$ implies $\int_X f \, d\mu \leq \int_X g \, d\mu$: every simple function admissible for f is also admissible for g. For $c \geq 0$, the same definition gives $\int_X cf \, d\mu = c \int_X f \, d\mu$. For $c > 0$, use the correspondence $\phi \mapsto c\phi$ between the two classes of simple functions; the case $c = 0$ follows from our convention.

Example 15 (Infinite values and integrals)**.** *Let m be Lebesgue measure on $\mathbb{R}$, and let C be the middle-thirds Cantor set.*

(i) *The function $f = +\infty\,\mathbf{1}_C$ has integral zero. Indeed, if $0 \le \phi \le f$ is simple, all its positive level sets are contained in the null set C, so $\int \phi\, dm = 0$.*

(ii) *For $f = +\infty\,\mathbf{1}_{[0,1]}$, the simple function $N\mathbf{1}_{[0,1]}$ lies below f for every $N \in \mathbb{N}$. Consequently $\int f\, dm \ge N$ for every N, so $\int f\, dm = +\infty$.*

(iii) *Define $f(x) = x^{-2}$ for $0 < |x| \le 1$, set $f(0) = +\infty$, and set $f(x) = 0$ for $|x| > 1$. For $0 < a < 1$,*

$$f \ge a^{-2}\mathbf{1}_{(0,a)}, \qquad \int_{\mathbb{R}} f\, dm \ge a^{-1}.$$

Thus $\int f\, dm = +\infty$, even though f is finite almost everywhere. The divergence comes from its values near zero, not merely from its value at zero.

(iv) *With the same f but with Dirac measure δ_0,*

$$\int_{\mathbb{R}} f\, d\delta_0 = f(0) = +\infty.$$

Whether a single point contributes to an integral depends on the measure.

9.2. **The monotone convergence theorem.**

Theorem 10 (Monotone convergence theorem)**.** *Suppose $f_n \in L^+$ and $f_n \uparrow f$ pointwise. Then $f \in L^+$ and*

$$\int_X f\, d\mu = \lim_{n\to\infty} \int_X f_n\, d\mu.$$

Both sides are allowed to be $+\infty$.

Proof. Measurability of f follows from Proposition 12. By monotonicity, the sequence of integrals is increasing and

$$A := \lim_{n\to\infty} \int_X f_n\, d\mu \le \int_X f\, d\mu.$$

To prove the reverse inequality, fix a simple function $0 \le \phi \le f$ and a number $0 < \alpha < 1$. Set

$$E_n = \{x \in X : f_n(x) \ge \alpha\phi(x)\}.$$

These sets are measurable: on each level set $\{\phi = a_j\}$, the defining condition is the measurable inequality $f_n \geq \alpha a_j$. Since f_n increases, $E_n \subseteq E_{n+1}$. In fact $E_n \uparrow X$. If $\phi(x) = 0$, then $x \in E_n$ for every n. If $\phi(x) > 0$, then

$$\lim_n f_n(x) = f(x) \geq \phi(x) > \alpha\phi(x),$$

so $x \in E_n$ for all sufficiently large n.

Now $\alpha\phi\mathbf{1}_{E_n} \leq f_n$, and therefore

$$\int_X f_n \, d\mu \geq \alpha \int_{E_n} \phi \, d\mu.$$

Apply continuity from below to the measure $\nu(E) = \int_E \phi \, d\mu$, established in Proposition 13. This gives

$$A \geq \alpha \lim_n \nu(E_n) = \alpha\nu(X) = \alpha \int_X \phi \, d\mu.$$

If $\int \phi \, d\mu = +\infty$, this already forces $A = +\infty$. Otherwise, let $\alpha \uparrow 1$ to obtain $A \geq \int \phi \, d\mu$. Taking the supremum over all such ϕ proves $A \geq \int f \, d\mu$. □

Remark 5. The factor $\alpha < 1$ is essential in this proof: $f_n(x)$ can approach $\phi(x) = f(x)$ from below without ever reaching it. It creates the strict gap needed to ensure $\bigcup_n E_n = X$.

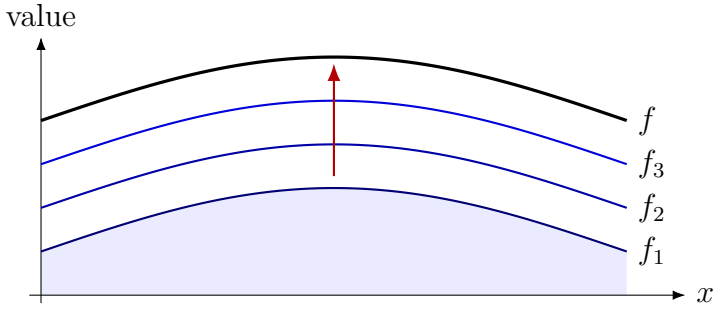


FIGURE 8. For an increasing sequence of nonnegative functions, the regions below the graphs increase to the region below the limit. The monotone convergence theorem makes this area intuition valid on an arbitrary measure space.

Corollary 5 (Computing with simple approximations)**.** *If $0 \le \phi_n \uparrow f$ are simple, then*

$$\int_X f \, d\mu = \lim_n \int_X \phi_n \, d\mu.$$

In particular, the approximations constructed in Theorem 9 compute the integral.

Proposition 14 (Integrating sums of nonnegative functions)**.** *If $f_n \in L^+$, then*

$$\int_X \left(\sum_{n=1}^{\infty} f_n \right) d\mu = \sum_{n=1}^{\infty} \int_X f_n \, d\mu.$$

The analogous identity holds for finite sums.

Proof. First take two functions $f, g \in L^+$. Choose nonnegative simple functions $\phi_n \uparrow f$ and $\psi_n \uparrow g$. Then $\phi_n + \psi_n \uparrow f + g$, so the monotone convergence theorem and additivity of the simple integral give

$$\begin{aligned} \int_X (f+g) \, d\mu &= \lim_n \int_X (\phi_n + \psi_n) \, d\mu \\ &= \lim_n \left(\int_X \phi_n \, d\mu + \int_X \psi_n \, d\mu \right) \\ &= \int_X f \, d\mu + \int_X g \, d\mu. \end{aligned}$$

All quantities are nonnegative, so no undefined subtraction occurs even when a limit is infinite. Induction proves finite additivity. Finally, apply monotone convergence to $S_N = \sum_{n=1}^{N} f_n \uparrow \sum_{n=1}^{\infty} f_n$:

$$\int_X \sum_{n=1}^{\infty} f_n \, d\mu = \lim_N \sum_{n=1}^{N} \int_X f_n \, d\mu = \sum_{n=1}^{\infty} \int_X f_n \, d\mu.$$

□

9.3. **Null sets and almost everywhere convergence.**

Lemma 5 (Zero integral)**.** *For $f \in L^+$,*

$$\int_X f \, d\mu = 0 \quad \iff \quad f = 0 \quad \mu\text{-a.e.}$$

Proof. If $f = 0$ a.e., every nonnegative simple function $\phi \le f$ vanishes a.e. Its positive level sets are null, so $\int \phi \, d\mu = 0$. Taking the supremum gives $\int f \, d\mu = 0$.

Conversely, suppose $\int f \, d\mu = 0$. For each $n \ge 1$, set $A_n = \{f > 1/n\}$. Since $f \ge n^{-1}\mathbf{1}_{A_n}$,

$$0 = \int_X f \, d\mu \ge \frac{1}{n}\mu(A_n),$$

whence $\mu(A_n) = 0$. The identity $\{f > 0\} = \bigcup_{n=1}^{\infty} A_n$ now proves the claim. □

Corollary 6 (Changing a function on a null set)**.** *If $f, g \in L^+$ and $f = g$ a.e., then $\int f \, d\mu = \int g \, d\mu$. More generally, $f \le g$ a.e. implies $\int f \, d\mu \le \int g \, d\mu$.*

Proof. Choose a measurable null set N outside which the equality or inequality holds. The lemma gives $\int f\mathbf{1}_N \, d\mu = \int g\mathbf{1}_N \, d\mu = 0$. Split each function into its parts on N and $X \setminus N$, use additivity, and compare the parts on $X \setminus N$. □

Corollary 7 (Almost everywhere monotone convergence)**.** *If $f_n, f \in L^+$ and $f_n \uparrow f$ a.e., then*

$$\lim_n \int_X f_n \, d\mu = \int_X f \, d\mu.$$

Proof. Outside one measurable null set N, both monotonicity and convergence hold. If these properties are initially stated separately a.e., take the union of their countably many exceptional null sets. Then $f_n\mathbf{1}_{X\setminus N} \uparrow f\mathbf{1}_{X\setminus N}$ everywhere. Apply Theorem 10 and Corollary 6. □

9.4. **Fatou's lemma.** Pointwise convergence by itself does not justify interchanging a limit and an integral.

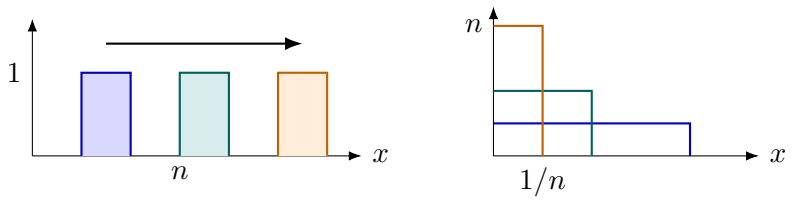
Example 16 (Two ways for mass to disappear)**.** *For Lebesgue measure on $\mathbb{R}$, consider*

$$u_n = \mathbf{1}_{[n,n+1]}, \qquad v_n = n\mathbf{1}_{(0,1/n)}.$$

Both sequences converge to zero at every point, but for every n,

$$\int u_n \, dm = \int v_n \, dm = 1.$$

In the first example the support moves to infinity; in the second it shrinks toward zero while the height increases. Neither sequence is increasing.



support moves to infinity height grows as support shrinks

FIGURE 9. Each rectangle has area 1, but both pointwise limits are zero. Endpoint values do not affect these Lebesgue integrals.

Theorem 11 (Fatou's lemma)**.** *For any sequence $f_n \in L^+$,*

$$\int_X \liminf_{n\to\infty} f_n \, d\mu \le \liminf_{n\to\infty} \int_X f_n \, d\mu.$$

In particular, if $f_n \to f$ a.e. and $f \in L^+$, then

$$\int_X f \, d\mu \le \liminf_{n\to\infty} \int_X f_n \, d\mu.$$

Proof. Let $g_k = \inf_{n\ge k} f_n$. These functions are measurable, nonnegative, and increase to $\liminf_n f_n$. Hence

$$\int_X \liminf_n f_n \, d\mu = \lim_k \int_X g_k \, d\mu$$

by monotone convergence. For every $n \ge k$, $g_k \le f_n$, so

$$\int_X g_k \, d\mu \le \inf_{n\ge k} \int_X f_n \, d\mu.$$

Letting $k \to \infty$ proves the inequality. In the almost everywhere convergence case, $\liminf_n f_n = f$ a.e.; changing a function on a null set does not change its integral. □

Example 17 (Fatou's lemma for series)**.** *On $\mathbb{N}$ with counting measure, the nonnegative integral is a sum: $\int h\,d\mu = \sum_{k=1}^{\infty} h(k)$. This follows by applying monotone convergence to $h\mathbf{1}_{\{1,\dots,N\}}$ (with truncation as well if h takes infinite values). Thus if $a_{n,k} \geq 0$ and $a_{n,k} \to a_k$ for each fixed k, then*

$$\sum_{k=1}^{\infty} a_k \leq \liminf_{n\to\infty} \sum_{k=1}^{\infty} a_{n,k}.$$

9.5. **Integrating real-valued functions.** Recall $f^+ = \max\{f, 0\}$ and $f^- = \max\{-f, 0\}$, so that $f = f^+ - f^-$ and $|f| = f^+ + f^-$. Both parts are nonnegative measurable functions whenever $f\colon X \to \mathbb{R}$ is measurable.

Definition 16 (Signed integral and integrability)**.** *If at least one of $\int f^+\,d\mu$ and $\int f^-\,d\mu$ is finite, define*

$$\int_X f\,d\mu = \int_X f^+\,d\mu - \int_X f^-\,d\mu.$$

If both are finite, f is called integrable*. Equivalently,*

$$f \text{ is integrable} \quad\Longleftrightarrow\quad \int_X |f|\,d\mu < \infty,$$

since $\int |f| = \int f^+ + \int f^-$. If both parts have infinite integral, $\int f$ is undefined: the expression $+\infty - (+\infty)$ has no assigned value.

Write $\mathcal{L}^1(X, \mu; \mathbb{R})$ for the real-valued integrable measurable functions. In the next lecture, L^1 will denote the space obtained by identifying functions that agree almost everywhere.

Proposition 15 (Linearity of the real integral)**.** *The set $\mathcal{L}^1(X, \mu; \mathbb{R})$ is a real vector space, and integration is a linear functional on it. Thus, for integrable f, g and $a, b \in \mathbb{R}$,*

$$\int_X (af + bg)\,d\mu = a\int_X f\,d\mu + b\int_X g\,d\mu.$$

Proof. First, $|af + bg| \leq |a||f| + |b||g|$, so $af + bg$ is integrable. If $c \geq 0$, then $(cf)^+ = cf^+$ and $(cf)^- = cf^-$, giving $\int cf = c\int f$. If $c < 0$, the positive and negative parts switch: $(cf)^+ = (-c)f^-$ and $(cf)^- = (-c)f^+$, giving the same identity.

For additivity, put $h = f + g$. The pointwise identity

$$h^+ + f^- + g^- = h^- + f^+ + g^+$$

involves only nonnegative functions. Integrating and using nonnegative additivity gives

$$\int h^+ + \int f^- + \int g^- = \int h^- + \int f^+ + \int g^+.$$

All six integrals are finite, so rearrangement is legitimate and yields $\int h = \int f + \int g$. □

Remark 6. If f, g are real-valued integrable functions and $f \leq g$ a.e., then $\int f\,d\mu \leq \int g\,d\mu$. Indeed, $g - f$ is nonnegative a.e., so its negative part has integral zero, and linearity gives $\int g - \int f = \int (g - f) \geq 0$.

10. September 24, 2026

We extend integration to complex-valued functions, explain why integrable functions are identified almost everywhere, and introduce the dominated convergence theorem. Throughout, $(X, \mathcal{M}, \mu)$ is a measure space.

10.1. **Complex-valued integrable functions.**

Definition 17. *A measurable function $f\colon X \to \mathbb{C}$ is* integrable *if* $\operatorname{Re} f$ *and* $\operatorname{Im} f$ *are integrable. Define*

$$\int_X f\,d\mu = \int_X \operatorname{Re} f\,d\mu + i \int_X \operatorname{Im} f\,d\mu.$$

Denote the space of these functions by $\mathcal{L}^1(X, \mu; \mathbb{C})$, or simply $\mathcal{L}^1(X, \mu)$.

The inequalities

$$|\operatorname{Re} f|,\ |\operatorname{Im} f| \leq |f| \leq |\operatorname{Re} f| + |\operatorname{Im} f|$$

show that, just as for real-valued functions,

$$f \in \mathcal{L}^1(X, \mu) \quad \Longleftrightarrow \quad \int_X |f|\,d\mu < \infty.$$

The space $\mathcal{L}^1(X, \mu)$ is a complex vector space, and its integral is complex linear. To see explicitly why complex scalars are allowed, write

$f = u + iv$ and $c = a + ib$. Then

$$cf = (au - bv) + i(bu + av),$$

and real linearity applied to both components gives $\int cf\, d\mu = c \int f\, d\mu$. Additivity follows componentwise. Also, $|af+bg| \leq |a||f|+|b||g|$ proves closure under complex linear combinations.

Example 18 (Lebesgue and Riemann integrability)**.**

(i) The characteristic functions $\mathbf{1}_{\mathbb{Q}}$ and $\mathbf{1}_C$ belong to $\mathcal{L}^1(\mathbb{R}, m)$ and have integral zero, where C is the Cantor set. On $[0, 1]$, $\mathbf{1}_{\mathbb{Q}}$ is not Riemann integrable because every lower sum is zero and every upper sum is one.

(ii) Define $f(x) = \sin x/x$ for $x \neq 0$ and $f(0) = 1$. Its improper Riemann integral over $\mathbb{R}$ converges, but $f \notin \mathcal{L}^1(\mathbb{R}, m)$. For convergence of the positive tail, integration by parts gives, for $B > A \geq 1$,

$$\int_A^B \frac{\sin x}{x}\, dx = \Big[-\frac{\cos x}{x}\Big]_A^B - \int_A^B \frac{\cos x}{x^2}\, dx.$$

The boundary term has a limit and the remaining integral converges absolutely. The negative tail is the same by symmetry. On the other hand, for every integer $k \geq 1$,

$$\int_{k\pi}^{(k+1)\pi} \frac{|\sin x|}{x}\, dx \geq \frac{1}{(k+1)\pi} \int_{k\pi}^{(k+1)\pi} |\sin x|\, dx = \frac{2}{(k+1)\pi}.$$

Summing and using divergence of the harmonic series proves $\int_{\mathbb{R}} |f|\, dm = +\infty$. In fact the positive and negative parts both have infinite integral, so the signed Lebesgue integral is undefined.

(iii) The function $x^{-1/2}$ is Lebesgue integrable on $(0, 1)$, with

$$\int_{(0,1)} x^{-1/2}\, dm = \lim_{n\to\infty} \int_{1/n}^1 x^{-1/2}\, dx = 2.$$

Here monotone convergence justifies the limit of the truncated integrals. The function is unbounded near zero, so it has no

proper Riemann integral on $[0,1]$*, although its improper Riemann integral converges.*

(iv) *The Gaussian* e^{-x^2} *is Lebesgue integrable on* $\mathbb{R}$*. For* $|x| \geq 1$*,* $e^{-x^2} \leq e^{-|x|}$*, while on* $[-1,1]$ *it is bounded by* 1*. These bounds give a finite integral. It is also improperly Riemann integrable.*

In the elementary computations above, we use agreement of the Riemann and Lebesgue integrals for continuous functions on compact intervals, which follows by squeezing between their lower and upper step functions.

10.2. **The integral triangle inequality.**

Proposition 16 (Triangle inequality and its equality case)**.** *If* $f \in \mathcal{L}^1(X,\mu)$*, then*

$$\left|\int_X f\,d\mu\right| \leq \int_X |f|\,d\mu.$$

Equality holds if and only if there is a constant $c \in \mathbb{C}$ *with* $|c| = 1$ *such that* $f = c|f|$ *a.e.*

Proof. Put $z = \int_X f\,d\mu$. If $z = 0$, the inequality is immediate. In this case equality holds precisely when $\int |f|\,d\mu = 0$, or equivalently $f = 0$ a.e. by Lemma 5. Such a function satisfies $f = c|f|$ a.e. for any $|c| = 1$.

Suppose now that $z \neq 0$, and choose $\alpha = \overline{z}/|z|$. Then $|\alpha| = 1$ and $\alpha z = |z|$. Complex linearity and the pointwise inequality $\operatorname{Re}(\alpha f) \leq |\alpha f| = |f|$ give

$$|z| = \operatorname{Re}\left(\alpha \int_X f\,d\mu\right) = \int_X \operatorname{Re}(\alpha f)\,d\mu \leq \int_X |f|\,d\mu.$$

Equality holds precisely when the nonnegative integrable function $|f| - \operatorname{Re}(\alpha f)$ has integral zero, hence when $\operatorname{Re}(\alpha f) = |f|$ a.e. A complex number w satisfies $\operatorname{Re} w = |w|$ exactly when it is a nonnegative real number. Thus equality forces $\alpha f = |f|$ a.e., or $f = \overline{\alpha}|f|$ a.e.

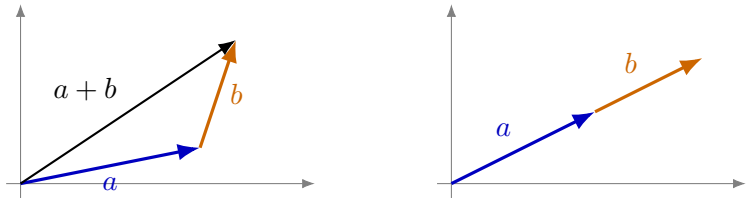
Conversely, if $f = c|f|$ a.e. for some $|c| = 1$, then $\int f = c\int |f|$, so taking absolute values gives equality. Here invariance of complex integrals under a.e. equality follows by applying Corollary 6 to the positive and negative parts of their real and imaginary components. □

For real-valued functions, equality means that $f \geq 0$ a.e. or $f \leq 0$ a.e. For complex-valued functions, all nonzero values must have the same phase, outside a null set.

Example 19. *Take $X = \{0, 1\}$ with counting measure, and set $f(0) = a$, $f(1) = b$. The integral triangle inequality becomes*

$$|a + b| \leq |a| + |b|.$$

Its equality case says that the two nonzero vectors point in the same direction.



$|a + b| < |a| + |b|$ $\qquad$ $|a + b| = |a| + |b|$

FIGURE 10. Different complex phases cause cancellation. Equality in the integral triangle inequality requires a common phase almost everywhere, as in the aligned vectors on the right.

10.3. **What integrals determine about a function.** A single integral cannot determine a function, since positive and negative contributions can cancel. Knowing the integral over every measurable set determines an integrable function up to a.e. equality.

Proposition 17. *For $f, g \in \mathcal{L}^1(X, \mu)$, the following are equivalent:*

(i) $f = g$ *a.e.;*

(ii) $\displaystyle\int_E f \, d\mu = \int_E g \, d\mu$ *for every* $E \in \mathcal{M}$*;*

(iii) $\displaystyle\int_X |f - g| \, d\mu = 0.$

Proof. The equivalence of (i) and (iii) is Lemma 5 applied to $|f - g|$. If (iii) holds, then for every $E \in \mathcal{M}$,

$$\left| \int_E f \, d\mu - \int_E g \, d\mu \right| \leq \int_E |f - g| \, d\mu \leq \int_X |f - g| \, d\mu = 0,$$

proving (ii).

Finally, suppose (ii) holds, and write $f - g = u + iv$. Taking real and imaginary parts shows that $\int_E u = \int_E v = 0$ for every E. For $A_n = \{u > 1/n\}$,

$$0 = \int_{A_n} u \, d\mu \geq \frac{1}{n}\mu(A_n),$$

so $\mu(A_n) = 0$. Taking the union over n proves $\mu(\{u > 0\}) = 0$. Apply the same argument to $-u$, v, and $-v$. Thus $u = v = 0$ a.e., giving (i). □

Remark 7 (Measurability after a null modification). A measurable modification on a null set leaves every integral unchanged. If the measure space is complete, arbitrary changes on a measurable null set preserve measurability. Without completeness, an arbitrary change need not be measurable; one can instead use a measurable extension, such as assigning zero on the exceptional set, or work with the completion of the measure.

10.4. **The space L^1.** If $E \in \mathcal{M}$, $\mu(X \setminus E) = 0$, and $f\colon E \to \mathbb{C}$ is measurable for the restricted σ-algebra, define its extension

$$\widetilde{f}(x) = \begin{cases} f(x), & x \in E, \\ 0, & x \notin E. \end{cases}$$

This extension is measurable. Define integrability and the integral of f using $\widetilde{f}$. Any other measurable extension gives the same integral, by Proposition 17. Thus a function need only be defined on a measurable set of full measure. Similarly, an extended-real-valued measurable function that is finite a.e. can be changed to zero where it is infinite. It is integrable precisely when the integral of its absolute value is finite.

Definition 18. *For $f, g \in \mathcal{L}^1(X, \mu)$, write $f \sim g$ if $f = g$ a.e. Define*

$$L^1(X, \mu) = \mathcal{L}^1(X, \mu)/\sim.$$

An element $[f] \in L^1(X, \mu)$ is an equivalence class of integrable functions. Equivalently, one can allow representatives defined only on measurable sets of full measure, comparing them on their common domain.

Set

$$\int_X [f]\,d\mu := \int_X f\,d\mu.$$

This is well defined by Proposition 17.

The relation $\sim$ is transitive because a finite union of null sets is null; reflexivity and symmetry are immediate. Vector operations $[f]+[g]=[f+g]$ and $c[f]=[cf]$ are independent of representatives for the same reason. We usually write $f \in L^1$ instead of $[f] \in L^1$, with the equivalence class understood.

Example 20 (Point evaluation is not generally well defined)**.** *For Lebesgue measure,*

$$[\mathbf{1}_{\mathbb{Q}}] = [0] \quad \text{in } L^1(\mathbb{R}, m),$$

although $\mathbf{1}_{\mathbb{Q}}(0) = 1$ *and the zero function takes value zero there. Thus an* $L^1(\mathbb{R}, m)$ *element has no intrinsic value at an individual point. This depends on the measure: with counting measure there are no nonempty null sets, so a.e. equality is pointwise equality.*

Proposition 18 (Completion does not change the quotient space)**.** *If* $(X, \overline{\mathcal{M}}, \overline{\mu})$ *is the completion of* $(X, \mathcal{M}, \mu)$*, then the natural map*

$$L^1(X,\mu) \longrightarrow L^1(X,\overline{\mu}), \qquad [f] \longmapsto [f],$$

is a bijection preserving integrals and integrals of absolute values.

Proof. An $\mathcal{M}$-measurable nonnegative function has the same integral for μ and $\overline{\mu}$: approximate it increasingly by $\mathcal{M}$-simple functions and apply monotone convergence to both measures. The corresponding assertion for integrable functions follows componentwise. Also, an $\mathcal{M}$-measurable set is $\overline{\mu}$-null exactly when it is μ-null, proving injectivity of the displayed map.

For surjectivity, let f be a complex-valued $\overline{\mathcal{M}}$-measurable integrable function. Choose $\overline{\mathcal{M}}$-simple functions $\phi_n \to f$ pointwise. By the description of the completion, each measurable set appearing in each ϕ_n differs from an $\mathcal{M}$-measurable set by a subset of an $\mathcal{M}$-measurable null set. Replace those sets to obtain $\mathcal{M}$-simple functions ψ_n agreeing with ϕ_n outside such null sets. There is one $N \in \mathcal{M}$, $\mu(N) = 0$, containing

all these exceptions, since only countably many sets were used. The functions $\psi_n \mathbf{1}_{X\setminus N}$ converge everywhere: to f outside N and to zero on N. Their limit g is $\mathcal{M}$-measurable and equals f $\overline{\mu}$-a.e. Thus g is integrable and represents the same class as f. □

Remark 8. The spaces of actual measurable functions can differ before taking the quotient. For example, a non-Borel subset A of the Cantor set is Lebesgue measurable and null, so $\mathbf{1}_A$ is Lebesgue measurable but not Borel measurable. Such subsets exist because the Cantor set has continuum cardinality, whereas there are only continuum many Borel sets. Nevertheless $\mathbf{1}_A$ represents the zero class in L^1.

Proposition 19 (The metric on L^1)**.** *The formula*

$$d([f],[g]) = \int_X |f-g|\,d\mu$$

defines a metric on $L^1(X,\mu)$. It is induced by the norm

$$\|[f]\|_1 = \int_X |f|\,d\mu.$$

Proof. Replacing either representative changes $|f-g|$ only on a null set, so d is well defined. It is finite because $|f-g| \le |f|+|g|$, and it is nonnegative and symmetric. Moreover,

$$d([f],[g]) = 0 \quad\Longleftrightarrow\quad f = g \text{ a.e.} \quad\Longleftrightarrow\quad [f]=[g].$$

The pointwise inequality $|f-h| \le |f-g|+|g-h|$ gives

$$d([f],[h]) \le d([f],[g]) + d([g],[h]).$$

Finally, $\|c[f]\|_1 = |c|\|[f]\|_1$, and the remaining norm axioms follow from the same arguments. □

On actual functions, the same formula is only a pseudometric: distinct functions can have distance zero. Passing to equivalence classes is exactly what makes the separation axiom valid.

10.5. **The dominated convergence theorem.** Monotone convergence requires an increasing nonnegative sequence. Fatou's lemma removes monotonicity but gives only one inequality. An integrable

bound permits passage to the limit even for complex functions and gives convergence in the L^1 metric.

Theorem 12 (Dominated convergence theorem)**.** *Let $f_n\colon X \to \mathbb{C}$ be measurable, and suppose $f_n \to f$ a.e., where f is measurable. Suppose there is a nonnegative $g \in \mathcal{L}^1(X,\mu)$ such that $|f_n| \le g$ a.e. for every n. Then $f_n, f \in \mathcal{L}^1(X,\mu)$ and*

$$\lim_{n\to\infty} \int_X |f_n - f|\, d\mu = 0.$$

Consequently,

$$\lim_{n\to\infty} \int_X f_n\, d\mu = \int_X f\, d\mu.$$

Proof. Take a common measurable null set outside which convergence and all the bounds hold, and redefine the functions to be zero there. This does not change their integrals. Thus we may assume $f_n \to f$ and $|f_n| \le g$ everywhere. Taking limits gives $|f| \le g$, so f and all f_n are integrable.

Put $h_n = |f_n - f|$. Then $h_n \to 0$ and $0 \le h_n \le 2g$. Apply Fatou's lemma to the nonnegative functions $2g - h_n$:

$$\begin{aligned} 2\int_X g\, d\mu &= \int_X \liminf_n (2g - h_n)\, d\mu \\ &\le \liminf_n \int_X (2g - h_n)\, d\mu \\ &= 2\int_X g\, d\mu - \limsup_n \int_X h_n\, d\mu. \end{aligned}$$

All integrals here are finite, and $0 \le \int h_n \le 2\int g$. Subtracting the finite quantity $2\int g$ yields $\limsup_n \int h_n \le 0$. Nonnegativity now implies $\int h_n \to 0$. Finally,

$$\left| \int_X f_n\, d\mu - \int_X f\, d\mu \right| \le \int_X |f_n - f|\, d\mu \longrightarrow 0.$$

□

Remark 9. If the pointwise limit is initially defined only on a measurable set of full measure, extend it by zero on the complement. It is then measurable as a pointwise limit of the similarly modified f_n.

Thus the theorem also applies naturally to a.e. defined representatives of L^1 elements.

Example 21 (Why the dominating function must be integrable)**.** *The smallest pointwise nonnegative bound for a sequence is $g_0 = \sup_n |f_n|$. Consider the following sequences on $\mathbb{R}$.*

(i) For $f_n = \mathbf{1}_{(n,n+1]}$, $n \geq 1$, we have $f_n \to 0$ pointwise and $\int f_n \, dm = 1$. But

$$g_0 = \mathbf{1}_{(1,+\infty)}, \qquad \int g_0 \, dm = +\infty.$$

Thus there is no integrable dominating function.

(ii) Let $f_n = n^{-1}\mathbf{1}_{(0,n]}$. Then $\sup_x |f_n(x)| = 1/n \to 0$, so the convergence to zero is even uniform, while $\int f_n \, dm = 1$ for every n. Here

$$g_0(x) = \begin{cases} 0, & x \leq 0, \\ 1/k, & k-1 < x \leq k, \quad k \in \mathbb{N}. \end{cases}$$

Indeed, for $k - 1 < x \leq k$, the nonzero terms are those with $n \geq k$, and their largest value is $1/k$. Therefore

$$\int_{\mathbb{R}} g_0 \, dm = \sum_{k=1}^{\infty} \frac{1}{k} = +\infty.$$

Uniform convergence on a space of infinite measure does not by itself imply convergence of integrals.

The same obstruction holds for a.e. domination: after taking the union of countably many exceptional null sets, any common bound dominates g_0 a.e., and hence also has infinite integral.

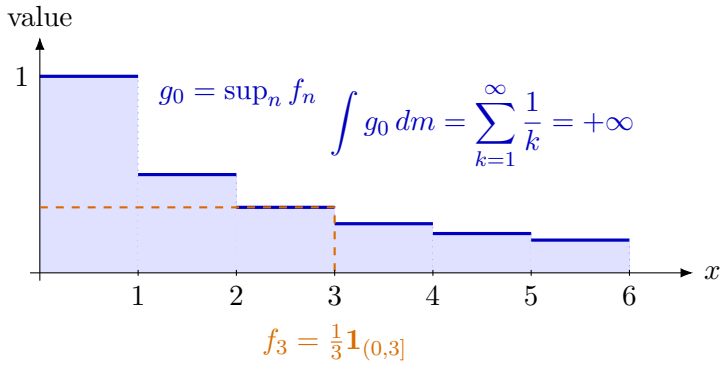


FIGURE 11. The envelope of $f_n = n^{-1}\mathbf{1}_{(0,n]}$ has a divergent harmonic tail. Every f_n is integrable, but no single integrable function dominates the sequence.

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Throughout, $(X, \mathcal{M}, \mu)$ is a measure space. We continue to write $\mathbf{1}_E$ for a characteristic function and m for Lebesgue measure. Recall that a measurable real- or complex-valued function is *integrable* precisely when

$$\int_X |f|\, d\mu < \infty.$$

We write $\mathcal{L}^1(X, \mu)$ for actual integrable functions and $L^1(X, \mu)$ for their equivalence classes under a.e. equality, usually denoting a class by its representative f. A nonnegative measurable function can have a well-defined integral equal to $+\infty$ without being integrable.

11.1. **Dominated convergence revisited.** Recall Theorem 12: if $f_n \to f$ a.e. and there is a nonnegative $g \in \mathcal{L}^1(X, \mu)$ such that $|f_n| \le g$ a.e. for every n, then f is integrable and

$$\int_X |f_n - f|\, d\mu \longrightarrow 0, \qquad \int_X f_n\, d\mu \longrightarrow \int_X f\, d\mu.$$

As before, a limit defined only a.e. is extended measurably by zero on a measurable null exceptional set.

Example 22. *On $X = (0, 1]$ with Lebesgue measure, let*

$$f_n(x) = \frac{e^{-x/n}}{\sqrt{x}}.$$

Then $f_n(x) \to x^{-1/2}$ *and* $0 \le f_n(x) \le g(x) = x^{-1/2}$. *Since* $\int_{(0,1]} g\,dm = 2$, *dominated convergence gives*

$$\lim_{n\to\infty} \int_{(0,1]} \frac{e^{-x/n}}{\sqrt{x}}\,dm = 2.$$

Since this sequence is increasing, Theorem 10 also gives the same conclusion by monotone convergence.

Alternative proof using Fatou's lemma. First suppose that all the functions are real-valued. After changing them on one measurable null set, assume that $f_n \to f$ and $|f_n| \le g$ everywhere. Then $|f| \le g$, so f is integrable. The functions $g + f_n$ and $g - f_n$ are nonnegative. Fatou's lemma (Theorem 11) applied to the first sequence gives

$$\begin{aligned}\int_X g\,d\mu + \int_X f\,d\mu &= \int_X \liminf_{n\to\infty} (g + f_n)\,d\mu \\ &\le \liminf_{n\to\infty} \int_X (g + f_n)\,d\mu \\ &= \int_X g\,d\mu + \liminf_{n\to\infty} \int_X f_n\,d\mu.\end{aligned}$$

Similarly,

$$\begin{aligned}\int_X g\,d\mu - \int_X f\,d\mu &\le \liminf_{n\to\infty} \int_X (g - f_n)\,d\mu \\ &= \int_X g\,d\mu - \limsup_{n\to\infty} \int_X f_n\,d\mu.\end{aligned}$$

Since $\int_X g\,d\mu$ is finite, subtraction is legitimate. Thus

$$\limsup_{n\to\infty} \int_X f_n\,d\mu \le \int_X f\,d\mu \le \liminf_{n\to\infty} \int_X f_n\,d\mu.$$

The reverse inequality between lim inf and lim sup always holds, so the limit exists and equals $\int_X f\,d\mu$.

For complex-valued functions, apply the real-valued result to the real and imaginary parts, both dominated in absolute value by g. Finally, applying the result to $|f_n - f| \le 2g$ also recovers convergence in L^1. $\square$

Example 23 (Why Fatou's lemma needs a lower bound)**.** *On* $\mathbb{R}$*, set* $f_n = -\mathbf{1}_{(n,n+1)}$*. Then* $f_n \to 0$ *pointwise, but*

$$\int_{\mathbb{R}} \liminf_n f_n \, dm = 0 > -1 = \liminf_n \int_{\mathbb{R}} f_n \, dm.$$

Likewise, for $h_n = \mathbf{1}_{(0,1)} - \mathbf{1}_{(n,n+1)}$*,*

$$h_n \longrightarrow \mathbf{1}_{(0,1)} \quad \text{pointwise}, \qquad \int h_n \, dm = 0, \qquad \int \liminf_n h_n \, dm = 1.$$

Thus Fatou's inequality cannot be applied directly to arbitrary signed functions. It does apply when there is a common integrable lower bound u*: apply the nonnegative version to* $f_n - u$ *and then add* $\int u$*. The proof above uses the lower bound* $-g$*.*

11.2. **Absolutely summable series of integrable functions.**

Corollary 8. *Suppose* $f_n \in \mathcal{L}^1(X, \mu)$ *and*

$$\sum_{n=1}^{\infty} \int_X |f_n| \, d\mu < \infty.$$

Then $\sum_{n=1}^{\infty} f_n(x)$ *converges absolutely for a.e.* x*. Its sum* f*, extended by zero on the exceptional set, is integrable, and*

$$\int_X f \, d\mu = \int_X \sum_{n=1}^{\infty} f_n \, d\mu = \sum_{n=1}^{\infty} \int_X f_n \, d\mu.$$

The partial sums also converge to f *in* L^1*.*

Proof. By Proposition 14, which follows from monotone convergence, the measurable function

$$g = \sum_{n=1}^{\infty} |f_n|$$

satisfies

$$\int_X g \, d\mu = \sum_{n=1}^{\infty} \int_X |f_n| \, d\mu < \infty.$$

In particular, $g < \infty$ a.e.: for every $M > 0$, $M\mu(\{g = +\infty\}) \leq \int g$, forcing this set to be null. Hence the series defining f converges absolutely a.e.

Set $F_N = \sum_{n=1}^{N} f_n$. Then $F_N \to f$ a.e. and $|F_N| \le g$. Changing g to zero on its null set of infinite values gives an integrable dominating function in the a.e. sense. Dominated convergence and finite linearity give

$$\int_X f\,d\mu = \lim_{N\to\infty} \int_X F_N\,d\mu = \lim_{N\to\infty} \sum_{n=1}^{N} \int_X f_n\,d\mu.$$

Dominated convergence also gives $\|F_N - f\|_1 \to 0$. More precisely, the pointwise tail estimate implies

$$\|f - F_N\|_1 \le \sum_{n=N+1}^{\infty} \|f_n\|_1 \longrightarrow 0.$$

Finally, Proposition 16 shows that the series of integrals is absolutely convergent as well. □

11.3. **Completeness of L^1.** Recall from Proposition 19 that

$$d(f,g) = \|f-g\|_1 = \int_X |f-g|\,d\mu$$

is a metric on the space of a.e. equivalence classes $L^1(X,\mu)$.

Theorem 13. *The metric space $(L^1(X,\mu), d)$ is complete. Equivalently, $L^1(X,\mu)$ with the norm $\|f\|_1 = \int_X |f|\,d\mu$ is a Banach space.*

Proof. Let (f_n) be Cauchy in L^1. Choose strictly increasing indices n_k such that

$$d(f_p, f_q) < 2^{-k} \qquad \text{whenever } p, q \ge n_k.$$

In particular, $d(f_{n_{k+1}}, f_{n_k}) < 2^{-k}$. Choose measurable representatives and define

$$g_1 = f_{n_1}, \qquad g_k = f_{n_k} - f_{n_{k-1}} \quad (k \ge 2).$$

Then

$$\sum_{k=1}^{\infty} \|g_k\|_1 \le \|f_{n_1}\|_1 + \sum_{k=2}^{\infty} 2^{-(k-1)} < \infty.$$

Corollary 8 gives an integrable function $f = \sum_{k=1}^{\infty} g_k$, with absolute convergence a.e. The partial sums telescope:

$$\sum_{k=1}^{K} g_k = f_{n_K}.$$

Thus $f_{n_K} \to f$ a.e. To see the metric convergence directly, set $h = \sum_{k=1}^{\infty} |g_k|$. We have $h \in L^1$ after a null modification, and

$$|f_{n_K} - f| \le 2h \quad \text{a.e.}$$

Dominated convergence gives $d(f_{n_K}, f) \to 0$.

Finally, convergence of this subsequence and the Cauchy property imply convergence of the original sequence. Given $\varepsilon > 0$, choose N so that $d(f_p, f_q) < \varepsilon/2$ for $p, q \ge N$, and then choose K with $n_K \ge N$ and $d(f_{n_K}, f) < \varepsilon/2$. For every $n \ge N$,

$$d(f_n, f) \le d(f_n, f_{n_K}) + d(f_{n_K}, f) < \varepsilon.$$

Hence $f_n \to f$ in L^1. □

Example 24 (Convergence in L^1 need not be pointwise)**.** *On* $[0,1]$ *with Lebesgue measure, list the characteristic functions of the dyadic intervals one level at a time. More precisely, for* $k \ge 0$ *and* $0 \le j < 2^k$*, set*

$$I_{k,j} = \begin{cases} [j2^{-k}, (j+1)2^{-k}), & j < 2^k - 1, \\ [1 - 2^{-k}, 1], & j = 2^k - 1, \end{cases} \qquad f_{2^k+j} = \mathbf{1}_{I_{k,j}}.$$

Thus the first levels are

$$\mathbf{1}_{[0,1]}; \qquad \mathbf{1}_{[0,1/2)},\ \mathbf{1}_{[1/2,1]}; \qquad \mathbf{1}_{[0,1/4)},\ \mathbf{1}_{[1/4,1/2)},\ \mathbf{1}_{[1/2,3/4)},\ \mathbf{1}_{[3/4,1]}.$$

The endpoint convention makes each level a partition of $[0,1]$*. For* $2^k \le n < 2^{k+1}$*,*

$$d(f_n, 0) = \int_{[0,1]} f_n \, dm = 2^{-k} \longrightarrow 0.$$

Nevertheless, each $x \in [0,1]$ *belongs to exactly one interval in every level. Thus* $f_n(x) = 1$ *infinitely often and* $f_n(x) = 0$ *infinitely often. In particular,*

$$\liminf_n f_n(x) = 0, \qquad \limsup_n f_n(x) = 1 \qquad (x \in [0,1]).$$

The sequence converges to zero in L^1 but converges pointwise nowhere. This explains why the completeness proof first selects a subsequence to obtain an a.e. limit.

Example 25 (An L^1 limit of continuous functions)**.** *On $\mathbb{R}$, define the continuous functions*

$$f_n(x) = \begin{cases} 0, & x \le -1/n, \\ 1 + nx, & -1/n < x < 0, \\ 1, & 0 \le x \le 1, \\ 1 - n(x-1), & 1 < x < 1 + 1/n, \\ 0, & x \ge 1 + 1/n. \end{cases}$$

Then $f_n \to \mathbf{1}_{[0,1]}$ pointwise and

$$\|f_n - \mathbf{1}_{[0,1]}\|_1 = \frac{1}{2n} + \frac{1}{2n} = \frac{1}{n} \longrightarrow 0.$$

Thus (f_n) is Cauchy in L^1, although its limit has no continuous representative. Indeed, any continuous representative would have to be identically zero on $(-\infty, 0)$ and identically one on $(0, 1)$, contradicting continuity at zero. Consequently, the subspace of continuous integrable functions is not complete in the L^1 metric.

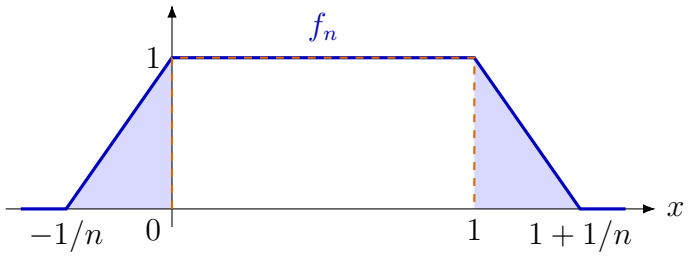


FIGURE 12. The two shaded triangles give the L^1 distance from f_n to $\mathbf{1}_{[0,1]}$. Their total area is $1/n$.

11.4. **Integrals depending on a parameter.** Let $-\infty < a < b < \infty$, and suppose $f\colon X \times [a, b] \to \mathbb{C}$ has integrable sections $f(\cdot, t)$ for

every $t \in [a,b]$. Define

$$F(t) = \int_X f(x,t)\, d\mu(x).$$

Theorem 14 (Continuity under the integral sign)**.** *Suppose there is a nonnegative $g \in \mathcal{L}^1(X,\mu)$ and a measurable null set N such that*

$$|f(x,t)| \le g(x) \qquad (x \notin N,\ t \in [a,b]).$$

If $t \mapsto f(x,t)$ is continuous at $t_0 \in [a,b]$ for every $x \notin N$, then F is continuous at t_0, relative to $[a,b]$.

Proof. For any sequence $t_n \to t_0$ in $[a,b]$, the functions $f_n(x) = f(x,t_n)$ converge a.e. to $f(x,t_0)$ and satisfy $|f_n| \le g$ a.e. Dominated convergence yields

$$F(t_n) = \int_X f(x,t_n)\, d\mu(x) \longrightarrow \int_X f(x,t_0)\, d\mu(x) = F(t_0).$$

The sequential criterion for continuity gives the result. □

Theorem 15 (Differentiation under the integral sign)**.** *Suppose there is a measurable null set N such that $t \mapsto f(x,t)$ is differentiable on (a,b) for every $x \notin N$. If there is a nonnegative $g \in \mathcal{L}^1(X,\mu)$ with*

$$\left|\frac{\partial f}{\partial t}(x,t)\right| \le g(x) \qquad (x \notin N,\ t \in (a,b)),$$

then F is differentiable on (a,b) and

$$F'(t) = \int_X \frac{\partial f}{\partial t}(x,t)\, d\mu(x).$$

The derivative in this integral is extended by zero on N.

Proof. Fix $t_0 \in (a,b)$ and an arbitrary sequence $t_n \to t_0$ with $t_n \in (a,b) \setminus \{t_0\}$. Put

$$h_n(x) = \frac{f(x,t_n) - f(x,t_0)}{t_n - t_0}.$$

Each h_n is measurable, and

$$h_n(x) \longrightarrow \frac{\partial f}{\partial t}(x,t_0) \qquad (x \notin N).$$

In particular, the derivative, extended by zero on N, is measurable as a pointwise limit of measurable functions; compare Proposition 12 and Corollary 4.

The mean value inequality gives $|h_n(x)| \le g(x)$ off N. For completeness, this remains valid for complex-valued f: choose $|\alpha| = 1$ so that $\alpha(f(x,t_n) - f(x,t_0))$ is nonnegative real, and apply the real mean value theorem to $t \mapsto \mathrm{Re}(\alpha f(x,t))$ on the interval with endpoints t_0, t_n. Its derivative has absolute value at most $g(x)$, proving the claimed inequality.

Dominated convergence and linearity now yield

$$\begin{aligned}\int_X \frac{\partial f}{\partial t}(x,t_0)\,d\mu(x) &= \lim_{n\to\infty} \int_X h_n\,d\mu \\ &= \lim_{n\to\infty} \frac{F(t_n) - F(t_0)}{t_n - t_0}.\end{aligned}$$

Since the sequence $t_n \to t_0$ was arbitrary, this is precisely differentiability at t_0 and the asserted formula for $F'(t_0)$. □

Remark 10. Both results are local in the parameter: it suffices to have the appropriate integrable bound on a neighborhood of the parameter under consideration. The exceptional null set must be common to all parameter values in that neighborhood.

Example 26 (Exponential integrals and factorials)**.** *Let $X = (0, +\infty)$ with Lebesgue measure, and set*

$$f(x,t) = e^{-tx}, \qquad F(t) = \int_0^\infty e^{-tx}\,dx = \frac{1}{t} \quad (t > 0).$$

For $1 \le t \le 2$,

$$\left|\frac{\partial f}{\partial t}(x,t)\right| = xe^{-tx} \le xe^{-x},$$

an integrable bound. Hence, for $1 < t < 2$,

$$-\frac{1}{t^2} = F'(t) = -\int_0^\infty xe^{-tx}\,dx.$$

More generally, on any compact parameter interval $[c,d]$ *with* $0<c<d$,

$$\left|\frac{\partial^r f}{\partial t^r}(x,t)\right| = x^r e^{-tx} \le x^r e^{-cx} \qquad (r\ge 0).$$

These bounds are integrable. Indeed, using the term of degree r *in the exponential series gives*

$$e^{cx/2} \ge \frac{(cx/2)^r}{r!}, \qquad x^r e^{-cx} \le r!\left(\frac{2}{c}\right)^r e^{-cx/2}.$$

Repeated differentiation under the integral sign is therefore justified on $(0,+\infty)$, *giving*

$$F^{(r)}(t) = \frac{(-1)^r r!}{t^{r+1}} = (-1)^r \int_0^\infty x^r e^{-tx}\,dx.$$

Consequently,

$$\int_0^\infty x^r e^{-tx}\,dx = \frac{r!}{t^{r+1}}, \qquad \int_0^\infty x^r e^{-x}\,dx = r! \qquad (r=0,1,2,\ldots).$$

To evaluate at $t=1$, *use a parameter interval containing* 1 *in its interior, for example* $[1/2,2]$.

11.5. **Lebesgue versus Riemann integration.** Let $f\colon [a,b]\to\mathbb{R}$ be bounded. For a partition

$$P=\{a=t_0<t_1<\cdots<t_n=b\},$$

define

$$m_j = \inf_{t\in[t_{j-1},t_j]} f(t), \qquad M_j = \sup_{t\in[t_{j-1},t_j]} f(t).$$

Definition 19 (Darboux sums and integrals)**.** *The* lower sum *and* upper sum *associated with* P *are*

$$s_P(f)=\sum_{j=1}^n m_j(t_j-t_{j-1}), \qquad S_P(f)=\sum_{j=1}^n M_j(t_j-t_{j-1}).$$

The lower and upper Riemann integrals are

$$\underline{I}_a^b(f)=\sup_P s_P(f), \qquad \overline{I}_a^b(f)=\inf_P S_P(f),$$

where P ranges over all partitions of $[a,b]$. The function f is Riemann integrable *when these two numbers agree; their common value is denoted by $\int_a^b f(x)\,dx$.*

Refining a partition increases its lower sum and decreases its upper sum. Taking a common refinement of any two partitions gives

$$\underline{I}_a^b(f) \le \overline{I}_a^b(f).$$

Equivalently, f is Riemann integrable if for every $\varepsilon > 0$ there is a partition P such that

$$S_P(f) - s_P(f) < \varepsilon.$$

Remark 11 (Connection with simple-function integration). The step functions with respective values m_j and M_j on $(t_{j-1}, t_j]$ bound f a.e. Their Lebesgue integrals are $s_P(f)$ and $S_P(f)$; endpoint values do not affect the integrals by Corollary 6. If f is Lebesgue measurable, it is integrable by boundedness, and

$$\underline{I}_a^b(f) \le \int_{[a,b]} f\,dm \le \overline{I}_a^b(f).$$

Thus the two integrals agree whenever f is also Riemann integrable.

12. October 1, 2026

We complete the comparison of Riemann and Lebesgue integration, then study convergence in measure. As before, m denotes Lebesgue measure, $\mathbf{1}_E$ denotes a characteristic function, and $d(f,g) = \|f-g\|_1 = \int_X |f-g|\,d\mu$ is the metric on $L^1(X,\mu)$.

12.1. **Darboux sums and refining partitions.** Let $f\colon [a,b] \to \mathbb{R}$ be bounded, where $a < b$. Recall that for a partition $P = \{a = t_0 < t_1 < \cdots < t_n = b\}$ we set

$$m_j = \inf_{[t_{j-1},t_j]} f, \qquad M_j = \sup_{[t_{j-1},t_j]} f,$$

and define the lower and upper sums by

$$s_P(f) = \sum_{j=1}^{n} m_j(t_j - t_{j-1}), \qquad S_P(f) = \sum_{j=1}^{n} M_j(t_j - t_{j-1}).$$

If Q refines P, then

$$s_P(f) \le s_Q(f) \le S_Q(f) \le S_P(f).$$

Taking common refinements gives

$$\underline{I}_a^b(f) := \sup_P s_P(f) \le \inf_P S_P(f) =: \overline{I}_a^b(f).$$

Equality is precisely Riemann integrability, and the common value is $\int_a^b f(x)\,dx$.

Lemma 6. *There is a sequence of partitions (P_k) such that $P_k \subseteq P_{k+1}$,*

$$\operatorname{mesh}(P_k) := \max_j (t_j - t_{j-1}) \longrightarrow 0,$$

and

$$s_{P_k}(f) \uparrow \underline{I}_a^b(f), \qquad S_{P_k}(f) \downarrow \overline{I}_a^b(f).$$

Proof. For each k, choose partitions A_k, B_k with

$$s_{A_k}(f) > \underline{I}_a^b(f) - \frac{1}{k}, \qquad S_{B_k}(f) < \overline{I}_a^b(f) + \frac{1}{k}.$$

Let D_k be the partition into 2^k equal subintervals, and let P_k be the common refinement of $A_1, \dots, A_k$, $B_1, \dots, B_k$, and D_k. These partitions are nested and have mesh at most $(b-a)2^{-k}$. The refinement inequalities and the choices of A_k, B_k give the asserted limits. □

12.2. **Riemann integrability and the discontinuity set.**

Theorem 16 (Lebesgue's criterion for Riemann integrability)**.** *Let $f\colon [a,b] \to \mathbb{R}$ be bounded.*

(i) If f is Riemann integrable, then it is Lebesgue measurable and integrable, and

$$\int_a^b f(x)\,dx = \int_{[a,b]} f\,dm.$$

(ii) The function f is Riemann integrable if and only if

$$D := \{x \in [a,b] : f \text{ is discontinuous at } x\}$$

has Lebesgue measure zero. Continuity at the endpoints is understood relative to $[a,b]$.

Proof of (i). For each partition P, define the step functions

$$g_P = f(a)\mathbf{1}_{\{a\}} + \sum_{j=1}^{n} m_j \mathbf{1}_{(t_{j-1},t_j]}, \qquad G_P = f(a)\mathbf{1}_{\{a\}} + \sum_{j=1}^{n} M_j \mathbf{1}_{(t_{j-1},t_j]}.$$

Then $g_P \le f \le G_P$ everywhere, and

$$\int_{[a,b]} g_P \, dm = s_P(f), \qquad \int_{[a,b]} G_P \, dm = S_P(f).$$

Choose the nested partitions of Lemma 6. Refinement gives $g_{P_k} \uparrow g$ and $G_{P_k} \downarrow G$ for bounded measurable functions g, G, with $g \le f \le G$. If $|f| \le M$, all these functions are bounded in absolute value by M. Dominated convergence on the finite interval therefore gives

$$\int_{[a,b]} g \, dm = \underline{I}_a^b(f), \qquad \int_{[a,b]} G \, dm = \overline{I}_a^b(f).$$

One can also apply monotone convergence to $g_{P_k} + M$ and $M - G_{P_k}$.

If f is Riemann integrable, these integrals agree. Thus $G - g \ge 0$ has integral zero, so $G = g$ a.e. by Lemma 5. The inequalities $g \le f \le G$ imply $f = g$ outside the measurable null set $\{G \ne g\}$. Completeness of Lebesgue measure makes f Lebesgue measurable, even though measurability was not assumed. Boundedness gives integrability, and a.e. equality yields

$$\int_{[a,b]} f \, dm = \int_{[a,b]} g \, dm = \int_a^b f(x)\, dx.$$

□

Proof of (ii). For $x \in [a,b]$, define the local lower and upper envelopes

$$h(x) = \lim_{\delta \downarrow 0} \inf_{\substack{y \in [a,b] \\ |y-x| \le \delta}} f(y), \qquad H(x) = \lim_{\delta \downarrow 0} \sup_{\substack{y \in [a,b] \\ |y-x| \le \delta}} f(y).$$

These limits exist: as the neighborhood shrinks, the infimum increases and the supremum decreases, and both stay bounded. Moreover,

$$h(x) \le f(x) \le H(x).$$

The function f is continuous at x if and only if $h(x) = H(x)$. Indeed, continuity forces both envelopes to equal $f(x)$; conversely, equality makes the supremum and infimum on a sufficiently small neighborhood arbitrarily close to $f(x)$.

Use the same partitions and step functions as above, and let

$$T = \bigcup_{k=1}^{\infty} P_k.$$

This is a countable set, hence $m(T) = 0$. We claim that

$$g(x) = h(x), \qquad G(x) = H(x) \qquad (x \notin T).$$

Fix $x \notin T$. For any $\delta > 0$, if $\mathrm{mesh}(P_k) < \delta$, the closed partition interval containing x lies in $[x - \delta, x + \delta]$. Hence

$$g_{P_k}(x) \ge \inf_{\substack{y \in [a,b] \\ |y-x| \le \delta}} f(y).$$

First letting $k \to \infty$, then $\delta \downarrow 0$, gives $g(x) \ge h(x)$.

Conversely, for fixed k we have $\mathrm{dist}(x, P_k) > 0$. If $0 < \delta < \mathrm{dist}(x, P_k)$, the neighborhood $[x-\delta, x+\delta]$ lies inside the partition interval containing x, so

$$g_{P_k}(x) \le \inf_{\substack{y \in [a,b] \\ |y-x| \le \delta}} f(y).$$

Letting $\delta \downarrow 0$ and then $k \to \infty$ gives $g(x) \le h(x)$. The same argument with suprema gives $G(x) = H(x)$.

Since $h = g$ and $H = G$ outside the null set T, completeness of Lebesgue measure implies that h, H are measurable. In particular, $D = \{H - h > 0\}$ is measurable. The integral identities proved above hold for any bounded f, without assuming Riemann integrability. Consequently,

$$\overline{I}_a^b(f) - \underline{I}_a^b(f) = \int_{[a,b]} (G - g)\, dm = \int_{[a,b]} (H - h)\, dm.$$

Since $H - h \geq 0$, the right-hand side is zero if and only if $H = h$ a.e., or equivalently $m(D) = 0$. The left-hand side is zero precisely when f is Riemann integrable. □

12.3. **Examples of different modes of convergence.**

Example 27. *On* $(\mathbb{R}, \mathcal{L}, m)$*, consider*

$$f_n = \frac{1}{n}\mathbf{1}_{(0,n)}, \qquad g_n = \mathbf{1}_{(n,n+1)}, \qquad h_n = n\mathbf{1}_{[0,1/n]}.$$

(i) $f_n \to 0$ *uniformly, since* $\sup_x |f_n(x)| = 1/n$.

(ii) $g_n \to 0$ *pointwise, since each fixed* x *eventually lies outside* $(n, n+1)$*. The convergence is not uniform, since* $\sup_x |g_n(x)| = 1$.

(iii) $h_n \to 0$ *a.e., but not at every point:* $h_n(0) = n$*, while* $h_n(x) = 0$ *eventually for every* $x \neq 0$.

Nevertheless, for every n*,*

$$\int_{\mathbb{R}} f_n \, dm = \int_{\mathbb{R}} g_n \, dm = \int_{\mathbb{R}} h_n \, dm = 1.$$

All three functions are nonnegative, so their L^1 *distances to zero equal* 1*. None of these sequences converges to zero in* L^1*. Compare Example 21.*

Example 28 (The typewriter sequence revisited)**.** *Use the dyadic intervals* $I_{k,j}$ *from Example 24, with the same endpoint convention, and write* $p_{2^k+j} = \mathbf{1}_{I_{k,j}}$ *on* $[0,1]$*, where* $k \geq 0$ *and* $0 \leq j < 2^k$*. Then*

$$\|p_{2^k+j}\|_1 = m(I_{k,j}) = 2^{-k} \longrightarrow 0.$$

Thus $p_n \to 0$ *in* $L^1([0,1], m)$*, although* $p_n(x)$ *fails to converge at every* $x \in [0,1]$*: it takes the values* 0 *and* 1 *infinitely often. Hence* L^1 *convergence does not imply a.e. convergence of the whole sequence.*

12.4. **Convergence in measure.** Let $(X, \mathcal{M}, \mu)$ be a measure space, and let f_n, f be measurable complex-valued functions on X.

Definition 20. *The sequence* (f_n) *is* Cauchy in measure *if, for every* $\varepsilon > 0$,

$$\mu\big(\{x : |f_n(x) - f_m(x)| \geq \varepsilon\}\big) \longrightarrow 0 \qquad (n, m \to \infty).$$

It converges in measure *to* f *if, for every* $\varepsilon > 0$,

$$\mu\big(\{x : |f_n(x) - f(x)| \geq \varepsilon\}\big) \longrightarrow 0 \qquad (n \to \infty).$$

Convergence in measure implies the Cauchy property, since

$$\{|f_n - f_m| \geq \varepsilon\} \subseteq \{|f_n - f| \geq \varepsilon/2\} \cup \{|f_m - f| \geq \varepsilon/2\}.$$

Example 29. *For the sequences in Example 27, both* f_n *and* h_n *converge to zero in measure. For a fixed* $\varepsilon > 0$, *the set* $\{|f_n| \geq \varepsilon\}$ *is empty once* $1/n < \varepsilon$, *while*

$$m(\{|h_n| \geq \varepsilon\}) = \frac{1}{n} \qquad (n \geq \varepsilon).$$

The typewriter sequence also converges to zero in measure, since for $0 < \varepsilon \leq 1$,

$$m(\{|p_{2^k+j}| \geq \varepsilon\}) = 2^{-k}.$$

However, (g_n) *is not even Cauchy in measure. If* $n \neq m$, *the intervals* $(n, n+1)$ *and* $(m, m+1)$ *are disjoint, so*

$$m(\{|g_n - g_m| \geq 1/2\}) = 2.$$

Thus pointwise convergence need not imply convergence in measure on a space of infinite measure.

Lemma 7 (Markov's inequality)**.** *If* $u \colon X \to \mathbb{C}$ *is measurable and* $\varepsilon > 0$, *then*

$$\mu(\{|u| \geq \varepsilon\}) \leq \frac{1}{\varepsilon} \int_X |u| \, d\mu.$$

The inequality is understood in the extended sense if the integral is infinite.

Proof. Set $E = \{|u| \geq \varepsilon\}$. Since $|u|\mathbf{1}_E \geq \varepsilon \mathbf{1}_E$, monotonicity gives

$$\varepsilon \mu(E) \leq \int_E |u| \, d\mu \leq \int_X |u| \, d\mu.$$

Divide by ε. □

Proposition 20. *If $f_n \to f$ in $L^1(X,\mu)$, then $f_n \to f$ in measure.*

Proof. For each $\varepsilon > 0$, Markov's inequality yields

$$\mu(\{|f_n - f| \geq \varepsilon\}) \leq \frac{1}{\varepsilon}\int_X |f_n - f|\, d\mu = \frac{1}{\varepsilon} d(f_n, f) \longrightarrow 0.$$

□

12.5. **Limits of sequences that are Cauchy in measure.**

Theorem 17. *Let (f_n) be a sequence of measurable functions $X \to \mathbb{C}$ that is Cauchy in measure. Then there is a measurable function $f\colon X \to \mathbb{C}$ such that $f_n \to f$ in measure. Moreover, some subsequence (f_{n_j}) converges to f a.e.*

Proof. Choose strictly increasing indices n_j such that

$$\mu(\{|f_p - f_q| \geq 2^{-j}\}) < 2^{-j} \qquad (p, q \geq n_j).$$

Set $g_j = f_{n_j}$ and define

$$E_j = \{|g_{j+1} - g_j| \geq 2^{-j}\}, \qquad F_k = \bigcup_{j=k}^{\infty} E_j.$$

Then

$$\mu(F_k) \leq \sum_{j=k}^{\infty} \mu(E_j) \leq \sum_{j=k}^{\infty} 2^{-j} = 2^{1-k}.$$

If $x \notin F_k$ and $i > j \geq k$, the triangle inequality gives

$$|g_i(x) - g_j(x)| \leq \sum_{\ell=j}^{i-1} |g_{\ell+1}(x) - g_\ell(x)| \leq \sum_{\ell=j}^{i-1} 2^{-\ell} < 2^{1-j}. \tag{7}$$

The set $F = \bigcap_{k=1}^{\infty} F_k$ is measurable and null, since $\mu(F) \leq \mu(F_k) \leq 2^{1-k}$ for every k. If $x \notin F$, then $x \notin F_k$ for some k, and (7) shows that $(g_j(x))$ is Cauchy in $\mathbb{C}$. By completeness of $\mathbb{C}$, we may define

$$f(x) = \begin{cases} \lim\limits_{j\to\infty} g_j(x), & x \notin F, \\ 0, & x \in F. \end{cases}$$

The measurable functions $g_j \mathbf{1}_{X\setminus F}$ converge everywhere to f, so f is measurable. In particular, $g_j \to f$ a.e. This construction does not require the measure space to be complete.

Letting $i \to \infty$ in (7), with $j = k$, gives

$$|f(x) - g_k(x)| \le 2^{1-k} \qquad (x \notin F_k).$$

For any fixed $\varepsilon > 0$, once $2^{1-k} < \varepsilon$,

$$\{|f - g_k| \ge \varepsilon\} \subseteq F_k, \qquad \mu(\{|f - g_k| \ge \varepsilon\}) \le 2^{1-k} \longrightarrow 0.$$

Thus the subsequence converges to f in measure as well.

Finally, pass back to the original sequence. Fix $\varepsilon, \eta > 0$. The Cauchy property gives N such that

$$\mu(\{|f_n - f_m| \ge \varepsilon/2\}) < \eta/2 \qquad (n, m \ge N).$$

Choose j with $n_j \ge N$ and $\mu(\{|g_j - f| \ge \varepsilon/2\}) < \eta/2$. For every $n \ge N$,

$$\{|f_n - f| \ge \varepsilon\} \subseteq \{|f_n - g_j| \ge \varepsilon/2\} \cup \{|g_j - f| \ge \varepsilon/2\},$$

so $\mu(\{|f_n - f| \ge \varepsilon\}) < \eta$. Since η was arbitrary, $f_n \to f$ in measure. $\square$

13. October 6, 2026

Throughout, $(X, \mathcal{M}, \mu)$ is a measure space, m denotes Lebesgue measure, and $\mathbf{1}_E$ denotes a characteristic function. We continue the study of convergence in measure, prove Egoroff's and Lusin's theorems, and begin the construction of product measures.

13.1. **Egoroff's theorem.** Recall the definition of convergence in measure. For measurable $f_n, f : X \to \mathbb{C}$, we have $f_n \to f$ in measure if, for every $\delta > 0$,

$$\mu(\{x : |f_n(x) - f(x)| \ge \delta\}) \longrightarrow 0.$$

Almost everywhere convergence need not imply convergence in measure: on $\mathbb{R}$, $f_n = \mathbf{1}_{(n,n+1)}$ converges pointwise to zero, but $m(\{|f_n| \ge 1/2\}) = 1$ for every n. On a finite measure space, however, a.e. convergence has a stronger consequence.

Theorem 18 (Egoroff's theorem)**.** *Suppose $\mu(X) < \infty$ and $f_n, f \colon X \to \mathbb{C}$ are measurable, with $f_n \to f$ a.e. For every $\varepsilon > 0$, there is $E \in \mathcal{M}$ such that $\mu(E) < \varepsilon$ and $f_n \to f$ uniformly on $X \setminus E$. That is, for every $\delta > 0$ there is N such that*

$$|f_n(x) - f(x)| < \delta \qquad (n \geq N,\ x \in X \setminus E).$$

Proof. Let Z be a measurable null set outside which $f_n \to f$. For $n, k \geq 1$, define

$$E_n(1/k) = \bigcup_{j=n}^{\infty} \{x \in X \setminus Z : |f_j(x) - f(x)| \geq 1/k\}.$$

For fixed k, these measurable sets decrease as n increases, and pointwise convergence on $X \setminus Z$ gives

$$\bigcap_{n=1}^{\infty} E_n(1/k) = \varnothing.$$

Since $\mu(X) < \infty$, continuity from above implies $\mu(E_n(1/k)) \to 0$. Choose n_k so that

$$\mu(E_{n_k}(1/k)) < \varepsilon 2^{-k}, \qquad E = Z \cup \bigcup_{k=1}^{\infty} E_{n_k}(1/k).$$

Countable subadditivity gives $\mu(E) < \varepsilon$. Given $\delta > 0$, choose k with $1/k < \delta$. If $x \notin E$ and $n \geq n_k$, then

$$|f_n(x) - f(x)| < 1/k < \delta.$$

The index n_k is independent of x, proving uniform convergence. □

Corollary 9. *If $\mu(X) < \infty$ and $f_n \to f$ a.e., then $f_n \to f$ in measure.*

Proof. Fix $\delta, \varepsilon > 0$. Egoroff's theorem gives a measurable set E with $\mu(E) < \varepsilon$ such that $f_n \to f$ uniformly on $X \setminus E$. For all sufficiently large n,

$$\{|f_n - f| \geq \delta\} \subseteq E, \qquad \mu(\{|f_n - f| \geq \delta\}) < \varepsilon.$$

Since ε is arbitrary, the desired limit is zero. □

13.2. **Approximation in L^1.** The next goal is Lusin's theorem: a measurable function on a finite interval is continuous after restriction to a suitable compact set whose complement has arbitrarily small measure. We first establish the approximation results needed in its proof.

Lemma 8 (Density of integrable simple functions)**.** *For every $f \in \mathcal{L}^1(X,\mu)$ and $\varepsilon > 0$, there is an integrable simple function ϕ such that*

$$\|f - \phi\|_1 = \int_X |f - \phi|\, d\mu < \varepsilon.$$

Hence integrable simple functions, viewed modulo a.e. equality, form a dense subspace of $L^1(X,\mu)$.

Proof. By Theorem 9, choose simple functions $\phi_n \to f$ pointwise with $|\phi_n| \le |f|$. Each ϕ_n is integrable, and

$$|\phi_n - f| \le 2|f| \in \mathcal{L}^1(X,\mu).$$

Dominated convergence gives $\|\phi_n - f\|_1 \to 0$. □

Lemma 9 (Approximation on the real line)**.** *Let μ be a Lebesgue–Stieltjes measure on $\mathbb{R}$. For every $f \in \mathcal{L}^1(\mathbb{R},\mu)$ and $\varepsilon > 0$:*

(i) There is a simple function $\psi = \sum_{j=1}^{N} a_j \mathbf{1}_{F_j}$, where each F_j is a finite union of bounded open intervals, such that $\|f - \psi\|_1 < \varepsilon$.

(ii) There is a continuous function $g\colon \mathbb{R} \to \mathbb{C}$ that vanishes outside a bounded interval and satisfies $\|f - g\|_1 < \varepsilon$.

In particular, the space $C_c(\mathbb{R})$ of continuous functions with compact support is dense in $L^1(\mathbb{R},\mu)$.

Proof. First we approximate sets of finite measure. If E is measurable and $\mu(E) < \infty$, regularity (Theorem 7) gives, for any $\eta > 0$, a compact $K \subseteq E$ and an open $U \supseteq E$ such that

$$\mu(E \setminus K) < \eta/2, \qquad \mu(U \setminus E) < \eta/2.$$

Cover K by finitely many bounded open intervals contained in U, and let F be their union. Then $K \subseteq F \subseteq U$, so

$$E \triangle F := (E \setminus F) \cup (F \setminus E) \subseteq U \setminus K, \qquad \mu(E \triangle F) < \eta.$$

Here one may take $F = \varnothing$ if $K = \varnothing$.

For (i), choose an integrable simple function $\phi = \sum_{j=1}^{N} a_j \mathbf{1}_{E_j}$ in standard form such that $\|f - \phi\|_1 < \varepsilon/2$. Discard the zero coefficient. If $\phi = 0$, there is nothing to prove. Otherwise, the E_j are disjoint and $a_j \neq 0$, so

$$|a_j|\mu(E_j) \leq \int_{\mathbb{R}} |\phi| \, d\mu < \infty.$$

Choose finite unions F_j of bounded open intervals with

$$\mu(E_j \triangle F_j) < \frac{\varepsilon}{2N|a_j|}.$$

Since $|\mathbf{1}_{E_j} - \mathbf{1}_{F_j}| = \mathbf{1}_{E_j \triangle F_j}$, the function $\psi = \sum_{j=1}^{N} a_j \mathbf{1}_{F_j}$ satisfies

$$\|f - \psi\|_1 \leq \|f - \phi\|_1 + \sum_{j=1}^{N} |a_j|\mu(E_j \triangle F_j) < \varepsilon.$$

For (ii), it suffices to approximate $\mathbf{1}_I$ for a bounded open interval $I = (a, b)$: each F_j is a finite disjoint union of such intervals. For n large enough that $2/n < b - a$, define

$$h_n(x) = \begin{cases} 0, & x \leq a, \\ n(x - a), & a < x < a + 1/n, \\ 1, & a + 1/n \leq x \leq b - 1/n, \\ n(b - x), & b - 1/n < x < b, \\ 0, & x \geq b. \end{cases}$$

Then $h_n \in C_c(\mathbb{R})$, $0 \leq h_n \leq \mathbf{1}_I$, and

$$\|\mathbf{1}_I - h_n\|_1 \leq \mu((a, a + 1/n)) + \mu((b - 1/n, b)) \longrightarrow 0.$$

Both intervals decrease to the empty set, and their measures are finite because Lebesgue–Stieltjes measures are finite on bounded intervals. Thus continuity from above applies. The ramps lie inside (a, b), so the approximation is valid even if μ has atoms at a or b.

Apply (i) with error $\varepsilon/2$, express the resulting ψ as a finite linear combination of interval characteristic functions, and replace each by a sufficiently accurate h_n. The resulting finite linear combination g

belongs to $C_c(\mathbb{R})$ and satisfies $\|\psi - g\|_1 < \varepsilon/2$. The triangle inequality gives $\|f - g\|_1 < \varepsilon$. □

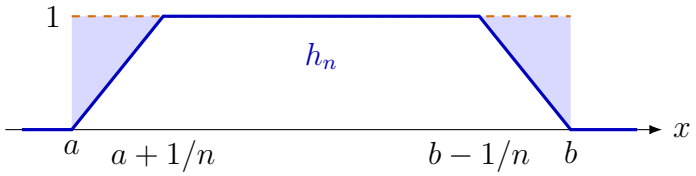


FIGURE 13. Continuous approximation to $\mathbf{1}_{(a,b)}$ from inside the interval. The error is supported near the endpoints.

13.3. **Lusin's theorem.**

Theorem 19 (Lusin's theorem)**.** *Let $f\colon [a,b] \to \mathbb{C}$ be Lebesgue measurable, where $-\infty < a < b < \infty$. For every $\varepsilon > 0$, there is a compact set $E \subseteq [a,b]$ such that*

$$m([a,b] \setminus E) < \varepsilon$$

and $f|_E$ is continuous in the relative topology on E.

Proof. Step 1: restrict to a compact set on which f is bounded. Since f is finite-valued,

$$A_N := \{x \in [a,b] : |f(x)| \geq N\} \downarrow \varnothing.$$

By continuity from above, $m(A_N) \to 0$. Fix N with $m(A_N) < \varepsilon/4$. By outer regularity, choose an open $U \supseteq A_N$ with $m(U) < \varepsilon/2$, and put $K = [a,b] \setminus U$. Then K is compact and $|f| < N$ on K. Extend $f|_K$ by zero to a function F on $\mathbb{R}$. It is measurable and integrable, since

$$\int_{\mathbb{R}} |F|\, dm \leq N(b-a) < \infty.$$

Step 2: obtain uniform convergence on a large subset. By Lemma 9, there are $g_j \in C_c(\mathbb{R})$ such that $\|F - g_j\|_1 \to 0$. Proposition 20 and Theorem 17 give a subsequence converging to F a.e. (limits in measure are unique a.e.). Relabel this subsequence as (g_j). On K, it converges

to f a.e. Egoroff's theorem on the finite measure space K gives a measurable $B \subseteq K$ such that

$$m(B) < \varepsilon/4, \qquad g_j \longrightarrow f \quad \text{uniformly on } K \setminus B.$$

Step 3: make the remaining set compact. Choose an open $V \supseteq B$ with $m(V) < \varepsilon/2$, and set $E = K \setminus V$. This is a closed subset of the compact set K, hence compact, and

$$m([a,b] \setminus E) \leq m(U) + m(V) < \varepsilon.$$

The restrictions $g_j|_E$ are continuous and converge uniformly to $f|_E$. A uniform limit of continuous functions is continuous, proving the assertion. □

Remark 12. Continuity of $f|_E$ refers only to points approaching within E; it does not assert that f is continuous on $[a,b]$ at every point of E. For example, $\mathbf{1}_{\mathbb{Q}\cap[a,b]}$ is discontinuous everywhere on $[a,b]$, but its restriction to any subset of the irrationals is identically zero.

13.4. **Product measures: the rectangle algebra.** Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be measure spaces. Recall from Lemma 1 that

$$\mathcal{M} \otimes \mathcal{N} = \sigma(\{A \times B : A \in \mathcal{M},\ B \in \mathcal{N}\}).$$

We seek a measure $\mu \times \nu$ on this product σ-algebra such that

$$(\mu \times \nu)(A \times B) = \mu(A)\nu(B),$$

with the convention $0 \cdot \infty = \infty \cdot 0 = 0$.

Proposition 21. *Let $\mathcal{A}$ consist of all finite disjoint unions of measurable rectangles $A \times B$. Then $\mathcal{A}$ is an algebra on $X \times Y$, and $\sigma(\mathcal{A}) = \mathcal{M} \otimes \mathcal{N}$.*

Proof. The empty set and $X \times Y$ are rectangles. Intersections of rectangles are rectangles, since

$$(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D).$$

Also, complements relative to $X \times Y$ have the disjoint decomposition

$$(A \times B)^c = (A^c \times B) \,\dot\cup\, (X \times B^c).$$

Any finite collection of rectangles can be refined into a finite disjoint collection: partition X according to membership in the sets in X and partition Y according to membership in the sets in Y, then take products of the resulting pieces. Thus finite unions, intersections, and complements of members of $\mathcal{A}$ belong to $\mathcal{A}$. The equality of generated σ-algebras follows because rectangles belong to $\mathcal{A}$ and every member of $\mathcal{A}$ is a finite union of rectangles. □

Remark 13. The collection of individual measurable rectangles is generally not an algebra: a union of two rectangles need not be a rectangle. Passing to finite unions is essential.

For a disjoint representation, define the candidate set function

$$\pi\left(\bigsqcup_{j=1}^{N} A_j \times B_j\right) = \sum_{j=1}^{N} \mu(A_j)\nu(B_j).$$

This is independent of the representation: refine two representations using the finite partitions of X and Y described above. Finite additivity of μ and ν shows that splitting a rectangle into these smaller rectangles preserves the sum. The same refinement argument shows that π is finitely additive on $\mathcal{A}$. The next step in constructing the product measure is to prove that π is a premeasure and apply the measure extension theorem.